

Quick Reference for Analyzing Reinsurance Contracts



Product Version: Touchstone Re 7.0
Publication Date: August 2019

Copyright

2019 AIR Worldwide Corporation. All rights reserved.

Information in this document is subject to change without notice. No part of this document may be reproduced or transmitted in any form, for any purpose, without the express written permission of AIR Worldwide Corporation (AIR).

Trademarks

AIR Worldwide, Touchstone, Touchstone Re, and CATRADER are registered trademarks of AIR Worldwide. Please refer to the following link for an up-to-date list of AIR trademarks: www.air-worldwide.com/Legal/Trademarks/.

In accordance with international patent conventions and treaties, if you use these trademarked items in any document you must also include the trademark symbol and prominently acknowledge AIR as the owner of the trademarks.

AIR Worldwide is a Verisk (Nasdaq:VRSK) business.



Internet Explorer and Windows Server are registered trademarks of Microsoft Corporation. SQL Server, Windows, and Excel are trademarks of Microsoft Corporation.

Confidentiality

AIR invests substantial resources in the development of its models, modeling methodologies, and databases. This document contains proprietary and confidential information and is intended for the exclusive use of AIR clients who are subject to the restrictions of the confidentiality provisions set forth in license and other nondisclosure agreements.

Contact Information

AIR is headquartered in Boston with additional offices in North America, Europe, and Asia. For information on our office locations, visit www.air-worldwide.com/About-AIR/Offices/.

AIR Worldwide Corporation
131 Dartmouth Street
Boston, MA 02116-5134
USA

Tel: (617) 267-6645
Fax: (617) 267-8284

If you need assistance in using the software or understanding the information in this document, please contact AIR Worldwide using the appropriate email:

- CATRADERSupport@air-worldwide.com for CATRADER questions
- TouchstoneSupport@air-worldwide.com for Touchstone questions
- TouchstoneReSupport@air-worldwide.com for Touchstone Re questions
- SoftwareDocumentationFeedback@air-worldwide.com for suggestions or questions regarding the documentation

Revision History

Revision Date	Description
August 2019	Updated figures: Removed figure 43 and replaced figure 44 with a new image.
June 2019	Converted to DITA
August 2018	Initial publication

Table of Contents

1 Introduction.....	9
2 Program Term Definitions.....	10
Retention and Limits.....	10
Applies to Areas and Event Filters.....	11
Percent Placed, Coinsurance and Participation Percentages.....	13
3 Catastrophe Excess of Loss.....	15
Creating Catastrophe Excess of Loss Program.....	15
Procedure.....	15
Contract Loss Examples.....	15
4 Quota Share.....	17
Creating Quota Share Programs.....	17
Contract Loss Examples.....	18
5 Drop-down Covers.....	19
Creating Drop-down Covers.....	19
Contract Loss Examples.....	20
6 Industry Loss Warranties.....	21
Creating Industry Loss Warranties.....	21
ILW Loss Examples.....	23
7 Parameter-based Industry Loss Warranties.....	25
Creating Parameter-based Industry Loss Warranties.....	25
Download UNICEDE/2 Take-Up Rate Import File.....	25
Import the Take-Up Rate File.....	26
Create a Program.....	27
8 Excess of Loss with (Outer) Aggregate Deductibles.....	31
Create Excess of Loss Programs with (Outer) Aggregate Deductibles.....	31
Inuring Program.....	31
Catastrophe Excess of Loss Program.....	31
Contract Loss Examples.....	33

9 Excess of Loss with Otherwise Recoverable (Inner Aggregate) Deductibles.....	34
Create Excess of Loss Programs with Otherwise Recoverable (Inner Aggregate) Deductibles.....	34
Contract Loss Examples.....	34
10 Franchise Deductible.....	36
Creating Per-Event Excess of Loss Programs with Franchise Deductibles.....	36
Contract Loss Examples.....	37
11 Reverse Franchise Deductible.....	38
Create Per-Event XOL Programs with Reverse Franchise Deductibles.....	38
Contract Loss Examples.....	39
12 Cascading Structures.....	40
Creating a Cascading Structure.....	40
Contract Examples.....	42
13 Top-and-Drop.....	44
Top-and-Drop Using Step Layers.....	44
Top-and-Drop Modeled as a Cascading Structure.....	44
Create a Top-and-Drop Structure.....	44
Contract Loss Examples.....	49
14 Nth-Event Covers.....	50
Creating Nth Event Covers.....	50
Second Occurrence Limit (Otherwise Recoverable).....	50
Second Event Exceeding Occurrence Retention.....	51
Contract Loss Examples.....	53

List of Figures

Figure 1. Occurrence and Aggregate Insurance Terms.....	10
Figure 2. Applies to Filters.....	11
Figure 3. Applies to Areas.....	12
Figure 4. Event Filter Tabs.....	13
Figure 5. Coinsurance and Participation Percentages.....	14
Figure 6. Layer Terms.....	15
Figure 7. Catastrophe Excess of Loss Contract Losses.....	16
Figure 8. Quota Share Program Terms.....	17
Figure 9. Applies to Areas.....	17
Figure 10. Quota Share Contract Losses.....	18
Figure 11. Layer 1 Program Terms.....	19
Figure 12. Step-Layer Program Terms.....	19
Figure 13. Drop-Down Cover Contract Losses.....	20
Figure 14. User Specified Market Shares.....	21
Figure 15. Florida Market Shares.....	22
Figure 16. ILW Program Terms.....	23
Figure 17. Company Loss Minimum.....	23
Figure 18. Industry Losses in Florida.....	24
Figure 19. Touchstone Re Company Data Import.....	26
Figure 20. Import UNICEDE/2 Take-Up Rates.....	27
Figure 21. ILW Program Terms.....	28
Figure 22. Add New Filter.....	28
Figure 23. Earthquake Magnitude Rule.....	29
Figure 24. Area Name Rule.....	29
Figure 25. Company Loss Minimum.....	30
Figure 26. Define Target.....	30
Figure 27. Inuring Program Terms.....	31

Figure 28. Catastrophe Excess of Loss.....	32
Figure 29. Inuring Programs.....	32
Figure 30. Selecting the relationship between programs.....	33
Figure 31. Excess of Loss Otherwise Recoverable Contract Losses.....	33
Figure 32. Per Event XL Excess of Amount Otherwise Recoverable.....	34
Figure 33. Catastrophe Excess of Loss with Outer Aggregate Deductible Contract Losses.....	35
Figure 34. Franchise Deductible Program Terms.....	36
Figure 35. Franchise Deductible Triggers.....	37
Figure 36. Franchise Deductible Contract Losses.....	37
Figure 37. Franchise Deductible Program Terms.....	38
Figure 38. Franchise Deductible Triggers.....	39
Figure 39. Reverse Franchise Deductible Contract Losses.....	39
Figure 40. CAT XOL Program for Each Individual Layer.....	40
Figure 41. Drag-and-Drop Programs Onto One Another to Define Relationship.....	41
Figure 42. Create Relationship Invoked Once Program Has Been Dropped onto Another.....	41
Figure 43. Dotted Line Indicates the Losses Are Inuring.....	42
Figure 44. Ground Up and Contract Loss for Each Layer Within Cascading Structure.....	43
Figure 45. Program Structure Showing Top-and-Drop Layer Structure.....	44
Figure 46. Top-and-Drop Program.....	45
Figure 47. CAT Layers Program.....	46
Figure 48. Parent Program (Top or Drop Layer) and Layers to be Sourced and/or Inured.....	47
Figure 49. Drag-and-Drop Program on top of another in the Relationships Pane to Invoke Create Relationship.....	47
Figure 50. Source Company Retention Layer: Drag-and-Drop Over the Catastrophe Layers.....	48
Figure 51. Three-Tier Structure with Company Retention Sourcing and Cat Layers Inuring to Top-and-Drop Program.....	48
Figure 52. Event Losses Flow Through Program Where First Event is Large Enough to Reach Limit of Top Layer.....	49
Figure 53. Event Losses Flow Through Program Where First Event Does Not Reach Program Limits So Top Layer Drops Down for Second Event.....	49
Figure 54. Nth Occurrence Limit (Otherwise Recoverable) Program Terms.....	51

Figure 55. Nth Event Exceeding Occurrence Retention Terms.....	52
Figure 56. Starting Event Number.....	52
Figure 57. Second Event Cover (Occurrence Limit) Contract Losses.....	53
Figure 58. Second Event Cover (Exceeding Occurrence Retention) Contract Losses.....	53

1 Introduction

This document is intended to serve as an easy reference for setting up a variety of contracts in Touchstone Re®. Also included are summaries of key program term definitions and how they are used by Touchstone Re. This document should be used as supplement to the Touchstone Re online help (available from the application or [AIR Client Portal](#)), which includes more complete descriptions of all fields and options available in the Touchstone Re user interface.

Only the most common contract types that the AIR Client Consulting and Support Group (CCSG) has received questions on in the past are covered in this document. New contract types will be added as we receive reader feedback. All questions and comments on improving this document can be sent to TouchstoneReSupport@air-worldwide.com.

This document describes how to create the following program types in Touchstone Re and analyze the results they generate:

- Catastrophe Excess of Loss
- Quota Shares
- Drop-Down Covers
- Industry Loss Warranties
- Parameter-based Industry Loss Warranties
- Excess of Loss with (Outer) Aggregate Deductibles
- Excess of Loss with Otherwise Recoverable (Inner Aggregate) Deductibles
- Franchise Deductibles
- Reverse Franchise Deductibles
- Cascading Structures
- Top-and-Drop Structures
- Nth Event Covers

2 Program Term Definitions

It is important to understand how the various fields on the Terms tab for a Touchstone Re program affect the behavior of the program and the assignment of losses to a contract. Several terms appear similar in function but actually behave very differently when determining contract losses.

Retention and Limits

Touchstone Re allows users to enter occurrence and aggregate terms for the loss retention and limit values used to model a contract. Occurrence terms dictate retention and limits on a per-occurrence or per-event basis, while aggregate terms dictate retention and limits on an annual basis (i.e., considering all events in a simulated year). Depending on the type of program selected, not all term fields will be available for use.

Program: ProgramABC

Program Details

Program Type: Inception Date: Expiration Date: Modified:

Aggregate Excess of Loss: 01/01/2019 12/31/2019 5/3/2019 by air-worldwide\52107

Currency Rate Table: Save Terms in: Display Terms in: Units:

AIR Default: US Dollar US Dollar Ones

Layers Program Options Relationships EP Adjustments

Terms

Layer	Treaty ID *	Occurrence Limit	Match Loss	Occurrence Retention	Aggregate Limit	Aggregate Retention	Multi-Year Reset Date	Applies to Areas	Applies to Events	Status	% Placed	In Coins
1	ID11595914		No					United States	All	Submitted	100%	
2	ID97971382							United States	All	Submitted	100%	

Figure 1. Occurrence and Aggregate Insurance Terms

The occurrence and aggregate terms are applied in the following order:

#	Terms	Definition
1	Occurrence Retention (Occ Ret)	For an event resulting in a company loss, the occurrence retention term is applied first. If the company loss is less than or equal to the occurrence retention term, all the loss is retained and none of the remaining terms are considered.
2	Occurrence Limit (Occ Limit)	If the loss exceeds the retention, Touchstone Re determines the eligible contract loss for that event by applying the occurrence limit term. Any loss exceeding the limit is not considered eligible and is retained.
3	Aggregate Retention (Agg Ret) ¹	Touchstone Re applies the aggregate retention term to the eligible contract loss, possibly reducing or eliminating the loss altogether. The retention value is reduced by the amount of eligible loss until it is completely exhausted by a single event or over multiple events.
4	Aggregate Limit (Agg Limit) ¹	After the application of the aggregate retention term, any remaining eligible contract loss results in an actual contract loss provided the aggregate limit for the year is not exceeded.

Applies to Areas and Event Filters

The **Applies to Areas** and **Applies to Events** allow you to apply filters to your Touchstone Re programs.

Program: ProgramABC

Program Details

Program Type: Inception Date: 01/01/2019 Expiration Date: 12/31/2019 Modified: 5/3/2019 by air-worldwide\j52107

Aggregate Excess of Loss: 01/01/2019

Currency Rate Table: Save Terms in: US Dollar Display Terms in: US Dollar Units: Ones

AIR Default: US Dollar

Layers | Program Options | Relationships | EP Adjustments

Terms

Layer	Treaty ID *	Occurrence Limit	Match Loss	Occurrence Retention	Aggregate Limit	Aggregate Retention	Multi-Year Reset Date	Applies to Areas	Applies to Events	Status
1	ID11595914		No					United States	All	Submitted
2	ID97971382									

Figure 2. Applies to Filters

¹ Aggregate terms are not available in Catastrophe Excess of Loss programs. However, you can control the aggregate limit by specifying the number of reinstatements on the layer in the **Reinstate** field. For example, a reinstatement factor of 0 would result in an aggregate limit equal to the occurrence limit.

The **Applies to Areas** dialog box allows you to restrict your program to a specific geographical area and, with pro rata treaties such as quota shares, to specify the ceded percentage on a country, area or subarea basis.

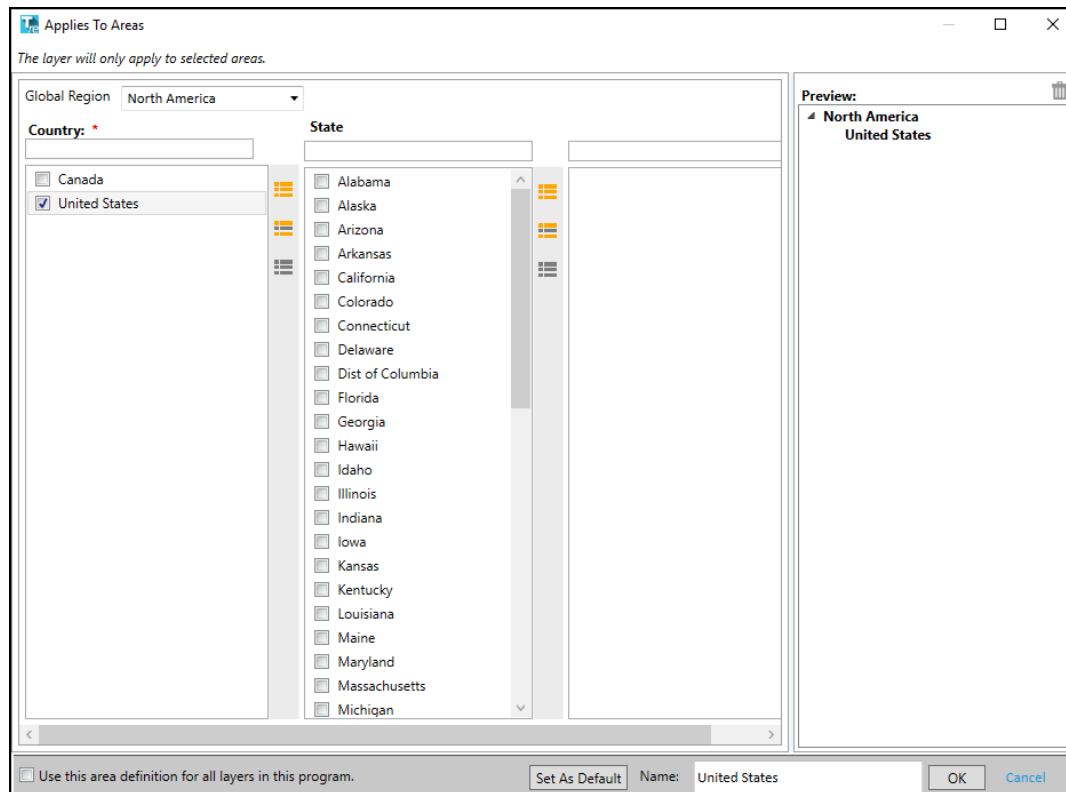


Figure 3. Applies to Areas

The ceded percentage determines the proportion of Company Loss assumed by the reinsurance treaty for that geographic area. The **Applies to Events** filter allows you to filter out events not meeting a user-specified criteria or rule. A variety of parameters are supported for earthquake and hurricane events on the **Rules** tab including:

• EQ Depth	• HU Max Landfall Windspeed	• HU Landfall Angle
• EQ Epicenter	• HU Central Pressure	• HU Landfall Criteria
• EQ Magnitude	• HU Carvill Hurricane Index SM (CME Hurricane Index) • HU Landfall Criteria	• HU Radius of Max Winds

Additionally, you can specify a geographic filter based on the epicenter or landfall location of the event. You can filter by a latitude/longitude polygon on the location by *Latitude/Longitude* tab or by specifying an area name, similar to the Applies to Areas filter, on the location *By Area* tab.

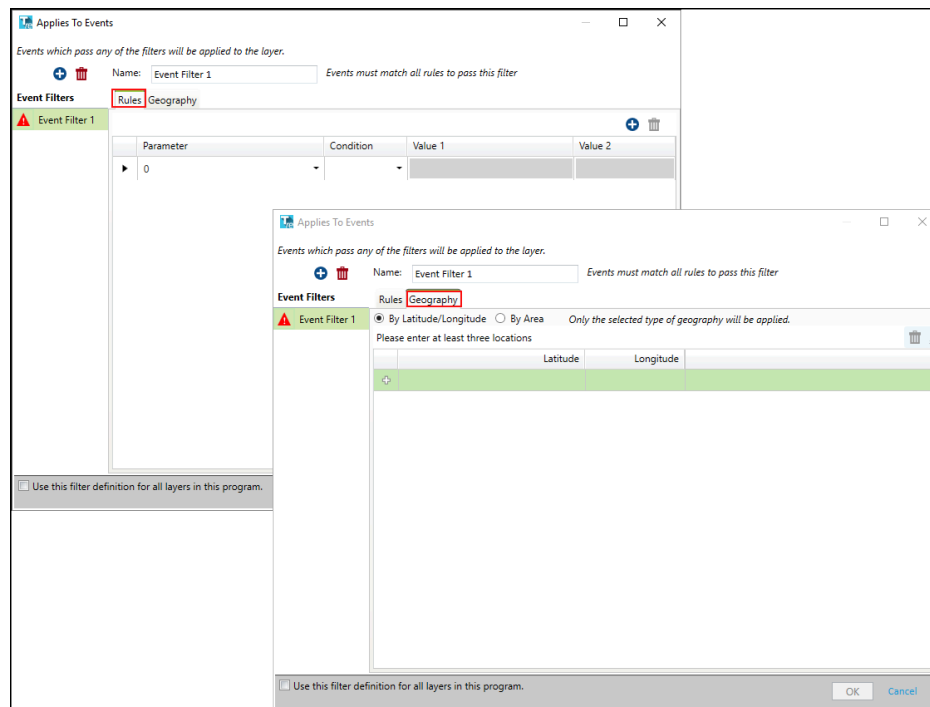


Figure 4. Event Filter Tabs

Events not meeting the criteria established in the in the event filter are not considered in the analysis.

Percent Placed, Coinsurance and Participation Percentages

Percent Placed is the percentage of the treaty that has been placed with reinsurers. If the treaty is not fully placed, then the insurer does not have coverage for that portion of the treaty and will absorb that portion of the losses as retained loss. Percent placed is not part of the reinsurance contract loss calculation. It does not affect 100%, Gross or Net participation contract loss. It is applied to determine the net retained loss only. Net retained loss = company loss – inuring program loss - (100% contract loss * % placed)

Insurer's Coinsurance is the portion of the losses to the layer that are retained by the cedant. If coinsurance is 10%, then the 10% of the layer loss is retained by the cedant and 90% is available to the reinsurer.

Gross participation, net participation, participation 3 and participation 4 fields can be used to reflect the individual reinsurer's share of each treaty. Therefore, a reinsurer that assumes 20% of a reinsurance contract would enter 20% in the gross participation field. The Net field allows them to reflect a different level of participation if a portion of their gross participation is being retroceded to another company. Participation 3 and 4 are optional user-defined participation percentages which allows the user to specify even more levels of retrocession. The 100% contract loss is unaffected by these terms; rather, your actual share of the contract loss is determined by scaling down the 100% contract loss by the gross, net participation, participation 3 or participation 4 rates. Each level of participation – 100%, gross, net, 3 and 4 – can be viewed on the Analysis pane.

Status	% Placed	Insurer's Coinsurance	Gross Participation	Net Participation	Participation 3	Participation 4
mitted ▼	100.00	0.00	0.00000	0.00000	0.00000	0.00000

Figure 5. Coinsurance and Participation Percentages

Note

The insurer's coinsurance field is enabled only for layered treaty structures, such as catastrophe excess of loss, aggregate excess of loss and per risk.

3 Catastrophe Excess of Loss

Catastrophe Excess of Loss contracts indemnify the ceding company for the portion of loss that exceeds the limit of a specified per-event retention.

Creating Catastrophe Excess of Loss Program

To create a three-layered catastrophe excess of loss program with 1 reinstatement and 95% coinsurance (insurer co-participating on 5% of each layer) with the following layer limits and retentions:

- **Layer 1** - \$50 million per occurrence limit in excess of \$50 million per occurrence retention
- **Layer 2** - \$75 million per occurrence limit in excess of \$100 million per occurrence retention
- **Layer 3** - \$100 million per occurrence limit in excess of \$175 million per occurrence retention

Procedure

1. From the Company Data pane select the Programs tab then click .
2. Select **Catastrophe Excess of Loss** from the **Add Program Type** dialog box then click **OK**.
3. Enter a unique name into the **Program Name** on the Terms tab.
4. Click **Add Layer** twice to create two new rows, one for the second layer in the program and one for the third.
5. Enter the program terms specified in the example above for each of the layers.
6. Click **Save** to store the newly created program.

Layers Program Options Relationships EP Adjustments													
Terms													
	Layer	Treaty ID *	Occurrence Limit	Match Loss	Occurrence Retention	Aggregate Limit	Aggregate Retention	Multi-Year Reset Date	Applies to Areas	Applies to Events	Status	% Placed	Insurer's Coinsurance
	1	ID11595914	50,000	No	50,000				United States	All	Submitted	100%	95%
	2	ID97971382	75,000	No	100,000				United States	All	Submitted	100%	95%
	3	ID82265648	100,000	No	175,000				United States	All	Submitted	100%	95%

Figure 6. Layer Terms

Contract Loss Examples

Using the program terms described in the previous section, Touchstone Re produces the contract losses for the company ground-up losses displayed below.

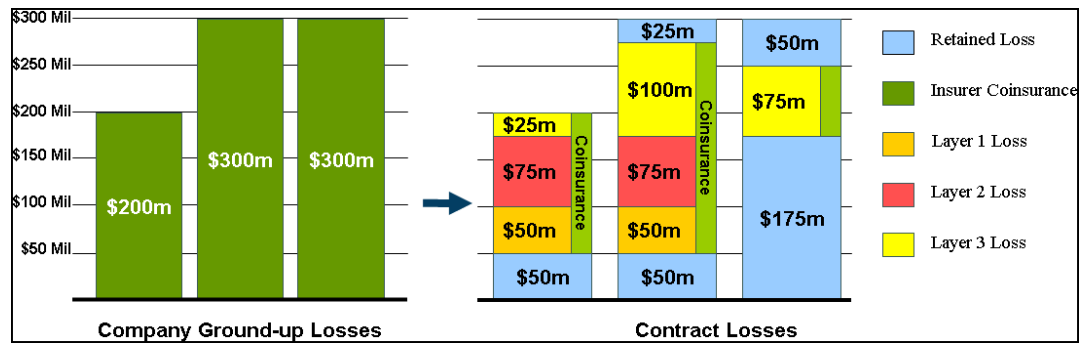


Figure 7. Catastrophe Excess of Loss Contract Losses

- **Event 1: Contract loss of \$142.5 M**
 - **Layer 1**– The first \$50 million is retained via the Layer 1 occurrence retention term, leaving \$150 million in loss, only \$50 million of which is eligible Layer 1 loss. The coinsurer absorbs 5% of the \$50 million, leaving \$47.5 million in losses.
 - **Layer 2**– \$75 million of the loss above the \$100 million layer retention is eligible Layer 2 loss. The coinsurer absorbs 5% of the \$75 million, leaving \$71.25 million in losses.
 - **Layer 3**– \$25 million of the loss above the \$175 million layer retention is eligible Layer 3 loss. The coinsurer absorbs 5% of the \$25 million, leaving \$23.75 million in losses.
- **Event 2: Contract loss of \$213.75 M**
 - **Layer 1**– Layer 1 uses its first and only reinstatement. The first \$50 million is retained via the Layer 1 occurrence retention term, leaving \$250 million in loss, only \$50 million of which is eligible Layer 1 loss. The coinsurer absorbs 5% of the \$50 million, leaving \$47.5 million in losses.
 - **Layer 2**– Layer 2 uses its first and only reinstatement. \$75 million of the loss above the \$100 million layer retention is eligible Layer 2 loss. The coinsurer absorbs 5% of the \$75 million, leaving \$71.25 million in losses.
 - **Layer 3**– Because this layer has 1 reinstatement, \$100 million of the loss above the \$175 million layer retention is eligible Layer 3 loss. The coinsurer absorbs 5% of the \$100 million, leaving \$95 million in losses. The amount above this is retained by the insurer.
- **Event 3: Contract loss of \$71.25 M**
 - **Layer 1** – Layer 1 has exhausted its reinstatement. No contract loss.
 - **Layer 2** – Layer 2 has exhausted its reinstatement. No contract loss.
 - **Layer 3** – Of the \$100 million of the loss above the \$175 million layer retention, only \$75 million is eligible under the contract. The layer limit was reinstated by event 2, leaving only a \$75 million of unused limit. The amount above this is retained by the insurer. The coinsurer absorbs 5% of the \$75 million, leaving \$71.25 million in losses.

4 Quota Share

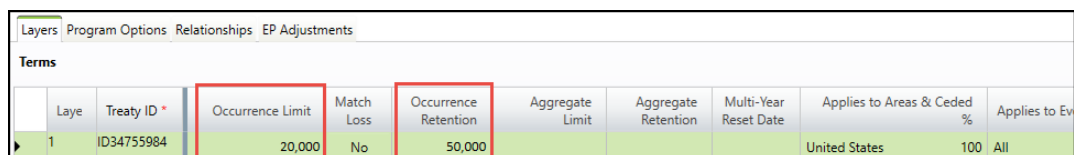
In a quota share contract, a defined percentage of company losses is ceded to the reinsurer on a proportional basis.

Creating Quota Share Programs

To create 20% a Quota Share program with \$20 million occurrence limit and a \$50 million aggregate limit:

Procedure

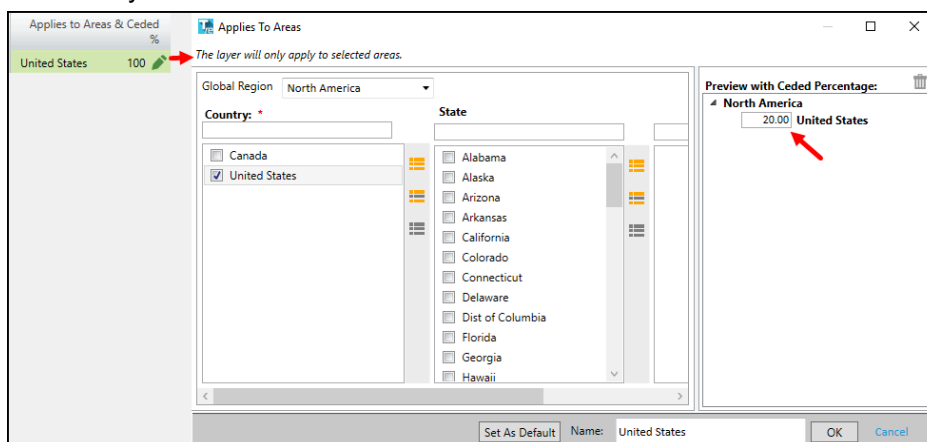
1. From the Company Data pane select the Programs tab then click .
2. Select **Quota Share** from the **Add Program Type** dialog box then click **OK**.
3. Enter a unique name into the **Program Name** on the Terms tab.
4. Enter the following program terms:
 - **Occ Limit:** \$20 million
 - **Agg Limit:** \$50 million



Layer	Treaty ID *	Occurrence Limit	Match Loss	Occurrence Retention	Aggregate Limit	Aggregate Retention	Multi-Year Reset Date	Applies to Areas & Ceded %	Applies to Ev
1	ID34755984	20,000	No	50,000				United States 100	All

Figure 8. Quota Share Program Terms

5. Click  corresponding to the Applies to Areas column to open the Applies to Areas dialog box.
6. Enter 20% in the Ceded % column for the U.S.
7. Click **OK** when you are finished.



Applies to Areas & Ceded %

United States 100 

The layer will only apply to selected areas.

Global Region: North America

Country: *

☐ Canada ☒ United States

State

Alabama, Alaska, Arizona, Arkansas, California, Colorado, Connecticut, Delaware, Dist of Columbia, Florida, Georgia, Hawaii

Preview with Ceded Percentage:

North America

20.00 United States

Set As Default Name: United States OK Cancel

Figure 9. Applies to Areas

8. Click **Save** to store the newly created program.

Contract Loss Examples

Using the program terms described in the previous section, Touchstone Re produces the contract losses for the company ground-up losses displayed here.

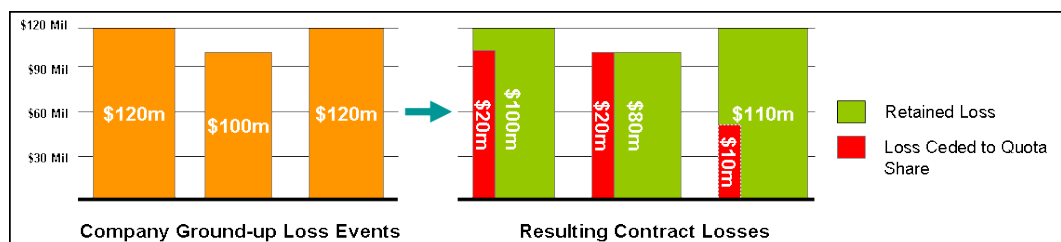


Figure 10. Quota Share Contract Losses

- **Event 1:** Contract loss of \$20 M

20% of the \$120 million company loss capped at the occurrence limit (\$20 million).

- **Event 2:** Contract loss of \$20 M

20% of the \$100 million company loss capped at the occurrence limit (\$20 million). After this event, \$10m of annual aggregate limit remains.

- **Event 3:** Contract loss of \$10 M

20% of the \$120 million company loss capped at the occurrence limit results in \$20 million of eligible loss under the per occurrence terms. The amount paid as contract loss is reduced to \$10 million due to the annual aggregate limit of \$50 million.

5 Drop-down Covers

Drop-down contracts allow insurers to lower retention for subsequent events occurring in a contract period after an initial occurrence retention or limit has been reached.

Creating Drop-down Covers

To create a drop-down cover contract in which the retention drops from \$60 million to \$30 million on the reinstatement of a layer:

Procedure

1. From the Company Data pane select the Programs pane then click .
2. Select **Catastrophe Excess of Loss** from the **Add Program Type** dialog box then click **OK**.
3. Enter a unique name into the **Program Name** on the Terms pane.
4. Enter the following program terms for Layer 1: Occ Limit: \$30 million
 - **Occ Limit:** \$30 million
 - **Occ Ret:** \$60 million

Layers Program Options Relationships EP Adjustments									
Terms									
	Layer	Treaty ID *	Occurrence Limit	Match Loss	Occurrence Retention	Applies to Areas	Applies to Events	Reinstatements	Sta
▶	1	ID17523589	30,000	No	60,000	United States	All		1 Subr

Figure 11. Layer 1 Program Terms

5. Click **Add Step-Layer**  to create a drop-down layer that will be activated after Layer 1 is exhausted.
6. Enter the following program terms for the new step-layer:
 - **Occ Limit:** \$30 million
 - **Occ Ret:** \$30 million

Layers Program Options Relationships EP Adjustments									
Terms									
	Layer	Treaty ID *	Occurrence Limit	Match Loss	Occurrence Retention	Applies to Areas	Applies to Events	Reinstatements	Sta
	1	ID17523589	30,000	No	60,000	United States	All		1 Subr
▶	2	ID20443243	30,000	No	30,000	United States	All		1 Subr

Figure 12. Step-Layer Program Terms

Contract Loss Examples

Using the program terms described in the previous section, Touchstone Re produces the contract losses for the company ground-up losses displayed below.

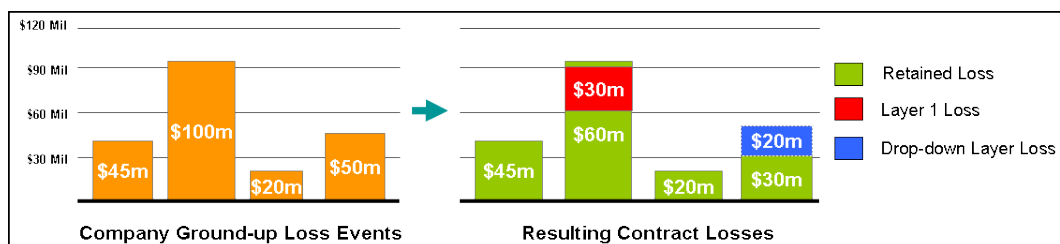


Figure 13. Drop-Down Cover Contract Losses

- **Event 1:** No contract loss

The loss falls below the occurrence retention of the primary layer (\$60 million).

- **Event 2:** Contract loss of \$30 M

The first \$60 million is retained via the occurrence retention term. The next \$30 million of loss is eligible contract loss and, as there are no aggregate terms, the full amount is paid as a contract loss. The primary layer is exhausted.

- **Event 3:** No contract loss

This loss falls below the occurrence retention of the step-layer (\$30 million).

- **Event 4:** Contract loss of \$20 M

The first \$30 million is retained via the occurrence retention. The next \$20 million is eligible contract loss and, as there are no aggregate terms, the full amount is paid as a contract loss.

6 Industry Loss Warranties

Industry Loss Warranties (ILWs) are based on the level of industry-insured loss incurred for a particular event. ILWs can cover a single peril, such as hurricane, or multiple perils.

Note



You can also apply parameter-based criteria, such as the location of hurricane landfall or earthquake epicenter, to Industry Loss Warranties. For more information on parameter-based ILWs, see the Chapter titled Parameter-Based Industry Loss Warranties.

Creating Industry Loss Warranties

To create a Florida-only \$10 billion Industry Loss Warranty with a \$5 million payment:

Procedure

1. Create a new company to serve as a proxy for the industry exposure in Florida.
2. In the Exposures pane, double-click on **United States** to open exposure detail.
3. Click the **Market Shares %** pane.
4. Select **Wind** from the Peril drop-down menu.
5. Select **User specified** from the Market Share Calculation drop-down menu.

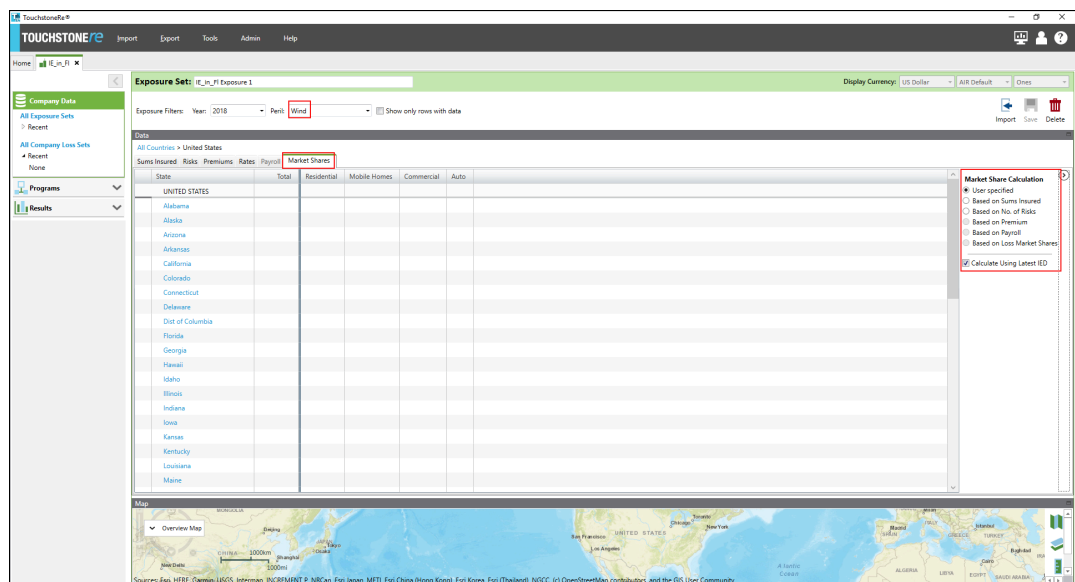


Figure 14. User Specified Market Shares

6. Set the market shares for each of the lines of business in Florida to 100%.
 - We assume a 100% insurance take-up rate for wind in the U.S.
 - For some countries and perils where not all properties are insured for a peril, industry exposure would be represented in Touchstone Re by entering market shares of less than 100%.

Data						
All Countries > United States						
Sums Insured Risks Premiums Rates Payroll Market Shares						
State	Total	Residential	Mobile Homes	Commercial	Auto	
UNITED STATES						
California						
Colorado						
Connecticut						
Delaware						
Dist of Columbia						
Florida		100.0000	100.0000	100.0000	100.0000	
Georgia						

Figure 15. Florida Market Shares

Note

To include other countries, states or counties in the Industry Loss Warranty, enter industry exposure for that region as User Specified market shares. AIR has posted recommended industry insurance take-up rates for all modeled perils and regions on the AIR Client Portal

7. Click **Save** to save your changes.
8. Click **Close** to return to the Company Detail screen.
9. From the Companies pane, select the Programs pane and then click **Add**.
10. Select Catastrophe Excess of Loss from the Add Program Type dialog box and then click **OK**.
11. Enter a unique name into the **Program Name** on the Terms pane.

12. Enter the following:

- **Exposure Set:** Use Specified Mkt Shrs
- **Use:** Specified Market Shares
- **Year:** Latest

Program: \$108 ILW

Program Details

Program Type: Catastrophe Excess of Loss Inception Date: 01/01/2018 Expiration Date: 12/31/2018 Modified: 12/26/2018

Currency Rate Table: AIR Default Save Terms in: US Dollar Display Terms in: US Dollar Units: Thousands

Define Target:

Before running an analysis, you must select a target for your program or associate a source program with it.

Company Loss Set: None Exposure Set: Florida Industry Market Use: Specified Market Shares Year: Latest

Loss Modification Factor: 1.000000

Figure 16. ILW Program Terms

13. To consider losses in Florida,

- Scroll down to the **Triggers** section.
- Enter a **Company Loss Minimum** of \$10 billion.

Note



Use Company Loss Minimum to consider losses in locations where the company has exposure. Use Industry Loss Minimum to consider losses for the entire U.S. and Caribbean for determining whether the trigger condition was met.

Triggers

Starting Event #: 1 Industry Loss Minimum: Company Loss Minimum: 10,000,000.00

Industry Loss Maximum: Company Loss Maximum:

Figure 17. Company Loss Minimum

14. Click **Save** to store the newly created program.

ILW Loss Examples

Using the program terms and triggers described in the previous section, Touchstone Re produces the Florida losses for the industry losses displayed here.

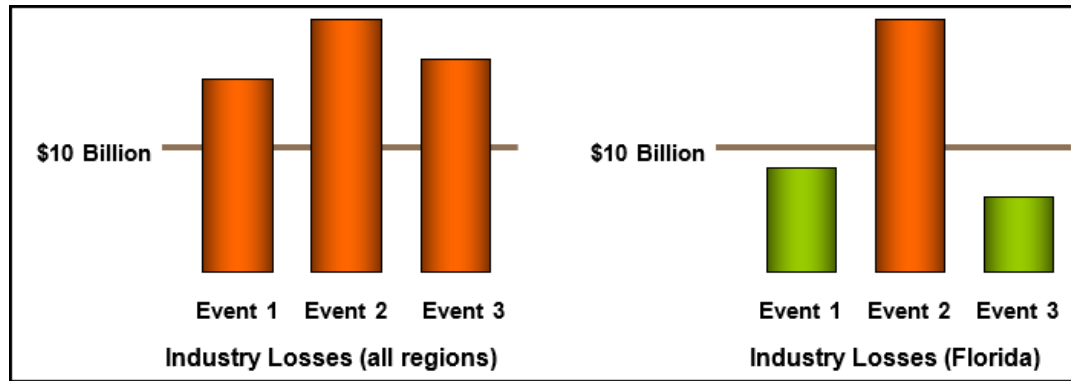


Figure 18. Industry Losses in Florida

All three events in exceed \$10 billion in industry losses. Only Event 2, however, exceeds \$10 billion in losses in Florida. Therefore, the program created in the previous section would only pay out the full \$5 million (Occ Limit) on Event 2, which meets the Company Loss Minimum of \$10 billion in Florida losses.

7 Parameter-based Industry Loss Warranties

Parameter-based ILWs are based on certain physical event parameters such as the location of hurricane landfall or epicenter and the intensity of the event. These ILWs can cover a single peril, such as hurricane, or multiple perils. Available parameters in Touchstone Re include:

- EQ Depth: Earthquake focal depth in kilometers.
- EQ Epicenter: Earthquake epicenter, where Inside refers to all earthquakes that occur within a country's border and Outside refers to all earthquakes that occur offshore or in a neighboring country.
- EQ Mw: Earthquake moment magnitude.
- HU Carvill Hurricane IndexSM: Carvill Hurricane Index (CME Hurricane Index) value, a continuous value which incorporates factors such as sustained wind speed and the radius of hurricane force winds to determine the potential for damage caused by the storm. For U.S. landfalling hurricanes only.
- HU Central Pressure: Hurricane central pressure at landfall in millibars.
- HU Max Landfall Windspeed: Maximum hurricane wind speed at landfall in miles per hour.
- HU Forward Speed: Hurricane forward speed.
- HU Landfall Angle: Hurricane landfall angle measured in degrees from North.
- HU Landfall Criteria: Events which make at least one landfall (Landfall) or none (Bypass)
- HU Radius of Max Winds: Radius of maximum winds.

Creating Parameter-based Industry Loss Warranties

To create a \$15 billion Industry Loss Warranty for earthquake events of magnitude 7.0 or greater with an epicenter in the seven state New Madrid region with a \$5 million contract payment you will:

- Download a current UNICEDE[®]/2 Take-Up Import Rate file.
- Import the Take-Up Rate file.
- Create a program.

Download UNICEDE/2 Take-Up Rate Import File

Procedure

1. Open a browser window and go to AIR Client Portal (www.air-worldwide.com/Support-and-Training/Client-Portal/).
2. Locate the most recent **UNICEDE/2 Take-Up Rate Import** posting, then double-click to open it.

- Download the zip file(s) associated with the posting per the instructions.

This UNICEDE/2 file simplifies the process of entering industry-insured exposures as User Specified market shares.

Import the Take-Up Rate File

To import the file(s) you downloaded in the previous section:

Procedure

- Create a new company to serve as a proxy for the industry exposure in the U.S.
- With the company open in the Company Data pane, select **Import** and then select UNICEDE / 2 from the **Import Type** drop-down.

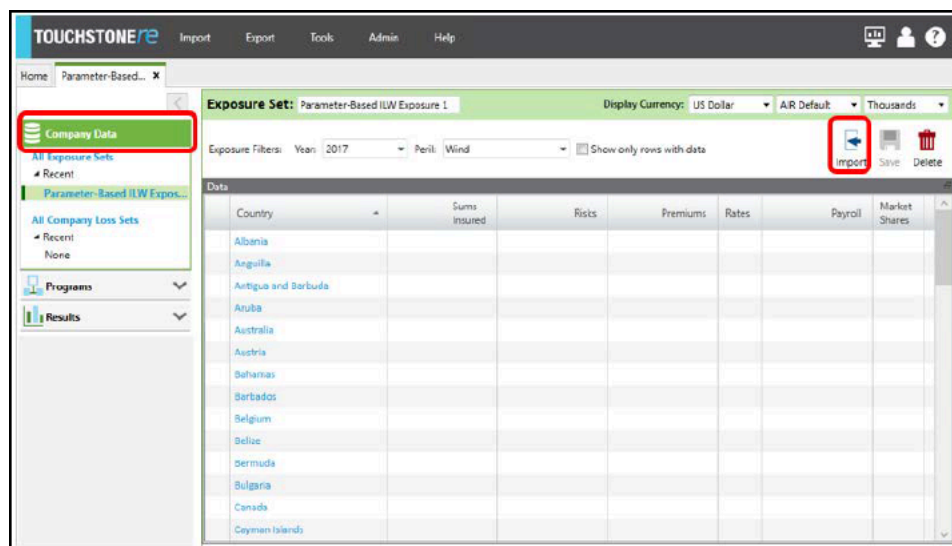


Figure 19. Touchstone Re Company Data Import

- Click **Browse** to locate and select the take-up rate file.

AIR ImportExpress

Import Type: UNICEDE®/2

Import data in the aggregate UNICEDE®/2 format.

Import Data | **Contact**

File Selection

Input File: * \\public\\unicede2\\uni_US.txt

Source

Exposure	SI	NR	PR	PY	MS
Wind	☒	☐	☐	☐	☒
Earthquake	☒	☐	☐	☒	☒
Fire	☒	☐	☐	☐	☒
Flood	☒	☐	☐	☐	☒
Terrorism	☐	☐	☐	☐	☐
MPCI (Agni)	☐	☐	☐	☐	☐
Crop Hail	☐	☐	☐	☐	☐
Pandemic	☐	☐	☐	☐	☐

Ceding Company: All_LOB_5
Company ID: 1000
Currency: USD
Units: Ones
Data as of: 01/25/2016

Currency Rate Table: AIR Default

Destination

Company: * Company B
Exposure Set Name: * Exposure 3
Import into Year: 2018

☒ Overwrite ☐ Append

☒ Close after Import

Verify Import Close

Figure 20. Import UNICEDE/2 Take-Up Rates

- Enable exposures from the Exposure table.
- Enter a unique name in Exposure Set Name.
- Click **Import**.
- Click **Yes** to update the area totals for the exposure that were imported.
- Click **Close** to return to the Company Data.

Note

The U.S. take-up rate file allows you to capture all industry insured losses caused by earthquakes occurring in the New Madrid region, regardless of the state in which they are incurred.

Create a Program

To create a program:

Procedure

- From the Company Data screen, select the **Programs** tab and then click .

2. Select Catastrophe Excess of Loss from the Add Program Type dialog box then click **OK**.
3. Enter a unique program name in the **Program Name** on the Terms tab.
4. Enter the following program terms:
 - Occ Limit: \$5 million
 - Reinsate: 1 (if applicable)

Layers								
Terms								
Layer	Treaty ID *	Occurrence Limit	Match Loss	Occurrence Retention	Applies to Areas	Applies to Events	Reinstatements	Status
1	ID35245731	5,000	No		United States	All	1	Submitted

Figure 21. ILW Program Terms

5. Click  corresponding to the Applies to Events .
6. In the Applies to Events dialog box, click  to create a new filter and define which events should be considered.

Program: New Madrid EQ >= 7.0

Program Details

Program Type: Catastrophe Excess of Loss
Inception Date: 01/01/2019
Expiration Date: 12/31/2019
Modified: 5/6/2019 by air-worldwide\52107
Currency Rate Table: Save Terms in: US Dollar
Display Terms in: US Dollar
Units: Ones

Terms

Layer	Treaty ID *	Occurrence Limit	Match Loss	Occurrence Retention	Applies to Areas	Applies to Events	Reinstatements	Status	% Plac
1	ID42070510	5,000.00	No. Dc		United States	All	1	Submitted	10

Applies to Events

Events which pass any of the filters will be applied to the layer.

Name: Event Filter 1

Event Filters

Parameter	Condition	Value 1	Value 2
CAN-Barley Price			
CAN-Barley Yield			
CAN MB UNSEED FRACTION			
CAN-MPCI Loss Type			
China MPCI			
EQ Depth (km)			
EQ Epicenter			
EQ Mw			
HU Carvill Hurricane Index			
HU Central Pressure (mb)			
HU Forward Speed (mph)			
HU Landfall Angle			
HU Landfall Criteria			
HU Max Landfall Windspeed (mph)			
HU Radius of Max Winds (miles)			
US-Corn Price			
US-Corn Yield			

Use this filter definition for all layers in this program.

OK Cancel

Figure 22. Add New Filter

7. On the Rules tab, enter a name for the new filter at the top of the Event Filter dialog box and configure the following magnitude rule:
 - Parameter: EQ Mw
 - Condition: >=
 - Value 1: 7.0

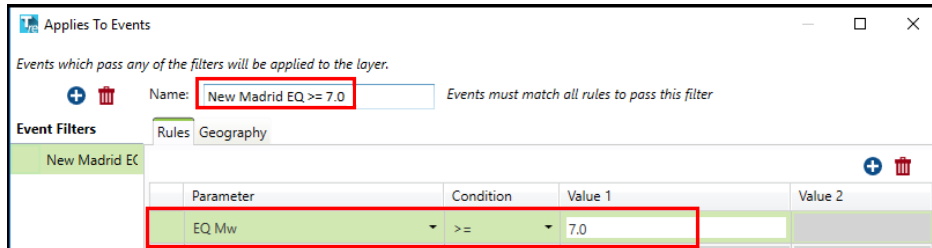


Figure 23. Earthquake Magnitude Rule

8. Click **OK**.
9. Select the **Geography** tab to set the location for the event.
10. Enable **By Area** from the Geography tab.
11. Select North America from the Regions drop-down menu.
12. Select the name of country and state.

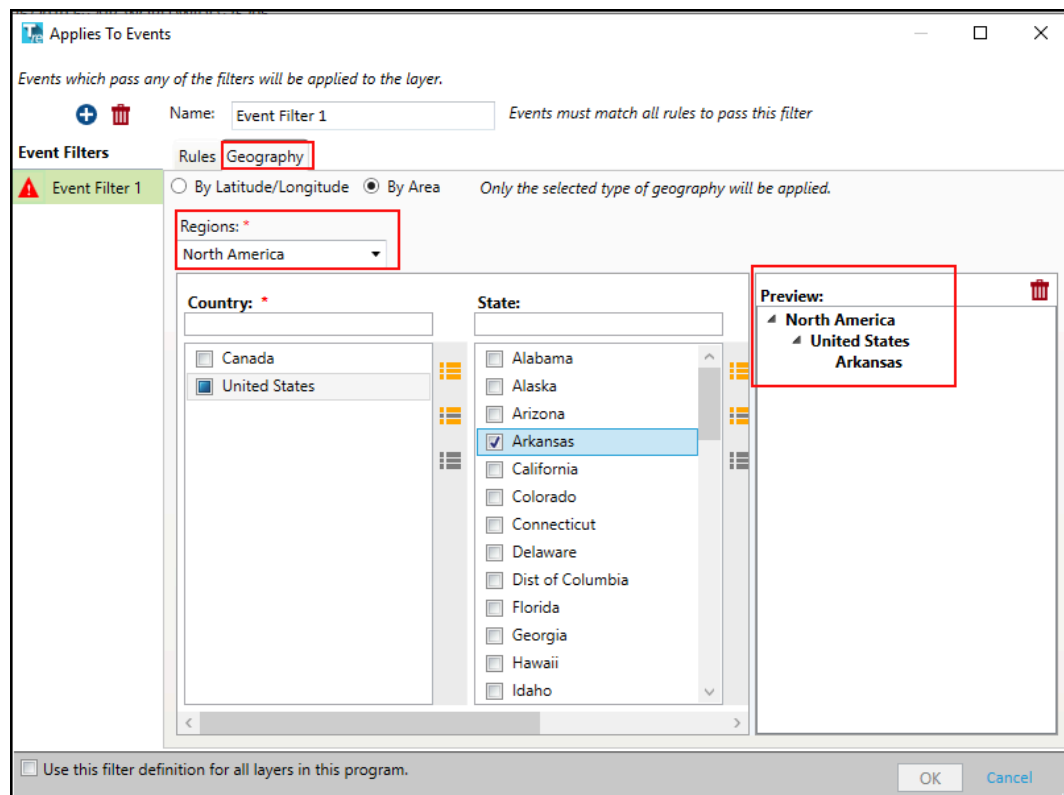


Figure 24. Area Name Rule

13. Scroll down to United States in the country list then double-click to open the state listing.

14. Toggle on the seven New Madrid states: Arkansas, Illinois, Indiana, Kentucky, Mississippi, Missouri, and Tennessee.
15. Click the Include label to display your selections at the top of the list.
16. Click **OK**.
17. Scroll down to the Triggers section and enter a Company Loss Minimum of \$15 billion.

Triggers			
Starting Event #:	1	Industry Loss Minimum:	
		Company Loss Minimum:	15,000,000.00
		Industry Loss Maximum:	
		Company Loss Maximum:	

Note



For this program, you should use the Company Loss Minimum to make Touchstone Re consider the county-level industry take-up rates (U.S. market shares). The Industry Loss Minimum considers state-level industry take-up rates.

Figure 25. Company Loss Minimum

18. Navigate to the **Program Options** tab and select the exposure set you created and use **Specified Market Shares**.

Program: New Madrid EQ >= 7.0				
Program Details				
Program Type:	Inception Date:	Expiration Date:	Modified:	
Catastrophe Excess of Loss	01/01/2018	12/31/2018	12/28/2018	
Currency Rate Table:	Save Terms in:	Display Terms in:	Units:	
AIR Default	US Dollar	US Dollar	Thousands	
Layers Program Options Relationships				
Define Target:				
Before running an analysis, you must select a target for your program or associate a source program with it.				
Company Loss Set:	None			
Exposure Set:	Parameter-Based ILW Ex			
Use:	Specified Market Shares	Year:	Latest	

Figure 26. Define Target

19. Click **Save** to store the newly created program.

8 Excess of Loss with (Outer) Aggregate Deductibles

Create this type of program when the aggregate deductible is to be applied prior to the per-occurrence terms.

Create Excess of Loss Programs with (Outer) Aggregate Deductibles

To create a program with a \$200 million occurrence limit and a \$100 million occurrence retention in excess of \$50 million aggregate loss (or outer aggregate), you must set up two programs:

- An inuring program
- A catastrophe excess of loss program

Inuring Program

The inuring program will capture the losses absorbed by the aggregate deductible. This is done as a separate program because aggregate terms are considered only after occurrence terms have been applied.

Procedure

1. From the Company Data, click the **Programs** tab and then click **Add**.
2. Select **Quota Share** from the **Add Program Type** dialog box and then click **OK**.

This program will inure to the benefit of the catastrophe excess of loss program.

3. Enter a unique name into the **Program Name** on the Terms tab.
4. Enter a \$50 million into the **Agg Limit**.

Layer	Treaty ID *	Occurrence Limit	Match Loss	Occurrence Retention	Aggregate Limit	Aggregate Retention	Multi-Year Reset Date	Applies to Areas & Ceded %	Applies to Events	Status
1	ID77971319		No		50,000			United States	100	All

Figure 27. Inuring Program Terms

5. Click **Save** to store the program.
6. Click **Close** to return to the Company Data pane.

Catastrophe Excess of Loss Program

To set up a catastrophe excess of loss program:

Procedure

1. From Company Data, select the **Programs** tab and then click [e](#).
2. Select Catastrophe Excess of Loss from the Add Program Type dialog box and then click **OK**.
3. Enter a unique name into the **Program Name** on the Terms tab.
4. Enter the following program terms:
 - Occurrence Limit: \$200 million
 - Occurrence Retention: \$100 million

Program: Cat XL Outer Agg

Program Details

Program Type: Catastrophe Excess of Loss
 Inception Date: 01/01/2019
 Expiration Date: 12/31/2019
 Modified: 5/6/2019 by air-worldwide\j52107

Currency Rate Table: AIR Default
 Save Terms in: US Dollar
 Display Terms in: US Dollar
 Units: Ones

Terms

Layer	Treaty ID *	Occurrence Limit	Match Loss	Occurrence Retention	Applies to Areas	Applies to Events	Reinstatements	Status
1	ID54620245	200,000.00	No. Dc	10,000.00	United States	All		1 Submit

Figure 28. Catastrophe Excess of Loss

5. Select the **Relationships** tab, then drag the quota share program created earlier "QS Inuring Program" (from the pane "Available Program Templates") onto the "Cat XL Outer Agg" program.

Program: Cat XL Outer Agg

Program Details

Program Type: Catastrophe Excess of Loss
 Inception Date: 01/01/2019
 Expiration Date: 12/31/2019
 Modified: 5/6/2019 by air-worldwide\j52107

Currency Rate Table: AIR Default
 Save Terms in: US Dollar
 Display Terms in: US Dollar
 Units: Ones

Relationships

Create Relationship

QS Inuring Program will be a child to Cat XL Outer Agg

Parent or Child: ☐ Parent ☒ Child

Relationship: Inuring

Perspective: Contract

Available Program Templates

Company	Assessment
New Madrid EQ...	1/1/2019-12/31/...
QS Inuring Progr...	1/1/2019-12/31/...

New Programs

Cat XOL	Quota Share
Agg XOL	Cat Bond
ILW	Ground Up

Relationship Properties Manager

Select any program on the stage to

Figure 29. Inuring Programs

6. In the **Create Relationships** dialog, select the inuring relationship, contract perspective.

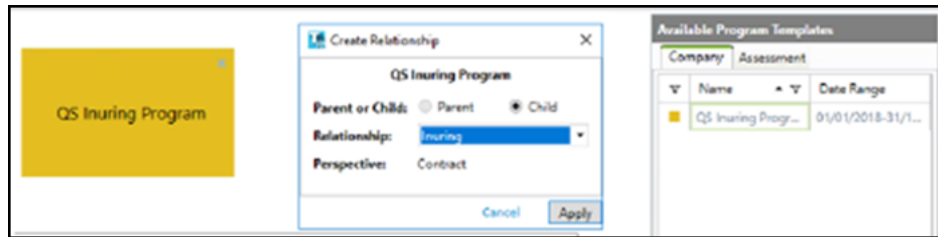


Figure 30. Selecting the relationship between programs

7. Click **Save** to store the newly created program.

Contract Loss Examples

Using the occurrence and aggregate terms described in the previous section, Touchstone Re produces the contract losses for the company ground-up losses displayed in the figure below.

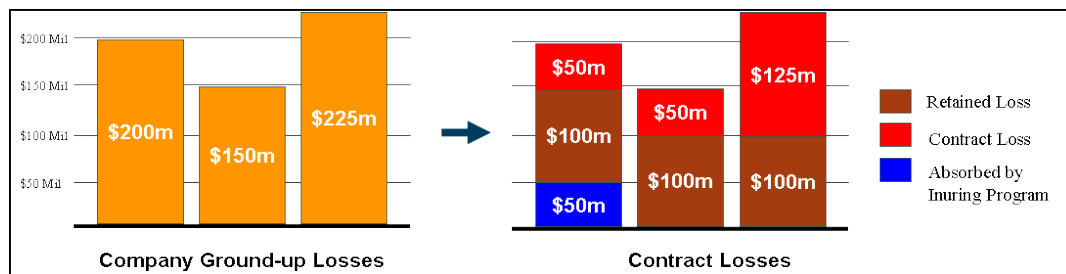


Figure 31. Excess of Loss Otherwise Recoverable Contract Losses

- **Event 1: Contract Loss of \$50 M**

The inuring quota share contract removes the first \$50 million in loss – outside of the occurrence terms – and is exhausted for the remainder of the contract period. The occurrence retention (\$100 million) is applied next. The remaining eligible loss is \$50 million, which is below the occurrence limit (\$200 million), and is paid as contract loss.

- **Event 2: Contract loss of \$50 M**

The first \$100 million is retained via the occurrence retention term. The remaining eligible loss of \$50 million is paid as a contract loss

- **Event 3: Contract loss of \$125 M**

The first \$100 million is retained via the occurrence retention term. The remaining eligible loss of \$125 million is paid in its entirety as a contract loss because there is a single reinstatement on the \$200 million occurrence limit, capping the aggregate limit at \$400 million for the year.

9 Excess of Loss with Otherwise Recoverable (Inner Aggregate) Deductibles

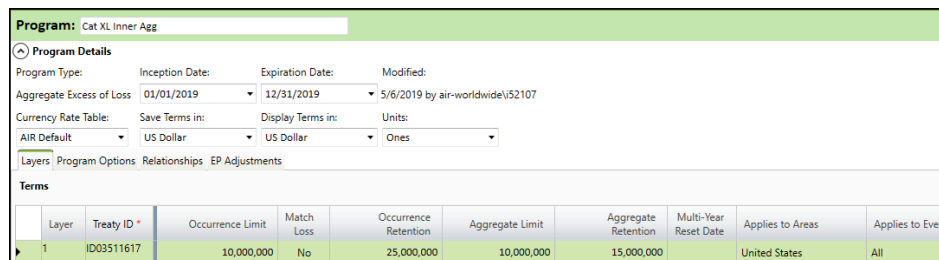
The otherwise recoverable amount, sometimes referred to as an inner aggregate deductible, absorbs some amount of loss that would qualify as contract losses if based solely on the per-occurrence retention and limit. The otherwise recoverable amount is not passed through as reinsured contract loss.

Create Excess of Loss Programs with Otherwise Recoverable (Inner Aggregate) Deductibles

To create a program with a \$10 million occurrence limit in excess of \$25 million with \$15 million otherwise recoverable (or inner aggregate):

Procedure

1. From the Company Data pane select the **Programs** tab and then click .
2. Select Aggregate Excess of Loss from the Add Program Type dialog box then click **OK**.
3. Enter a unique name into the **Program Name** on the Terms tab
4. Enter the following occurrence and aggregate terms:
 - Occ Limit: \$10 million
 - Occ Ret: \$25 million
 - Agg Limit: \$10 million
 - Agg Ret: \$15 million



Layer	Treaty ID *	Occurrence Limit	Match Loss	Occurrence Retention	Aggregate Limit	Aggregate Retention	Multi-Year Reset Date	Applies to Areas	Applies to Events
1	ID03511617	10,000,000	No	25,000,000	10,000,000	15,000,000		United States	All

Figure 32. Per Event XL Excess of Amount Otherwise Recoverable

5. Click **Save** to store the newly created program.

Contract Loss Examples

Using the program terms described in the previous section, Touchstone Re produces the contract losses for the company ground-up losses displayed in the figure below.

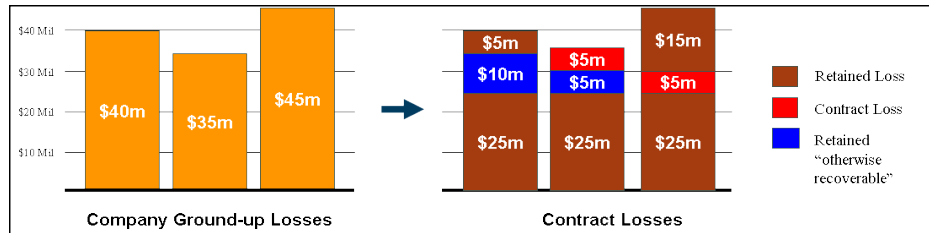


Figure 33. Catastrophe Excess of Loss with Outer Aggregate Deductible Contract Losses

- **Event 1: No contract loss**

The first \$25 million is retained via the occurrence retention term leaving \$15 million in loss, only \$10 million of which is eligible contract loss. The \$10 million in eligible contract loss is absorbed by the aggregate retention term (\$15 million), depleting the aggregate retention to \$5 million.

- **Event 2: Contract loss of \$5 M**

The first \$25 million is retained via the occurrence retention term leaving \$10 million in eligible contract loss. The first \$5 million in eligible contract loss is absorbed by the remaining aggregate retention (\$5 million) – exhausting it – and the remaining \$5 million is paid as a contract loss.

- **Event 3: Contract loss of \$5 M**

The first \$25 million is retained via the occurrence retention term leaving \$10 million in eligible contract loss. However, only the first \$5 million can be paid because of the aggregate limit (\$10 million) cap for the year.

10 Franchise Deductible

A franchise deductible, otherwise known as a “disappearing deductible,” is a per-occurrence loss threshold that must be reached before a contract loss will be paid. Unlike a typical deductible, however, the ultimate reinsured loss is not reduced by the franchise deductible amount.

Creating Per-Event Excess of Loss Programs with Franchise Deductibles

To create a per-event excess of loss program with \$300 million occurrence limit and a \$100 million franchise deductible:

Procedure

1. From the Company Data screen select the Programs tab then click [e](#).
2. Select Catastrophe Excess of Loss from the Add Program Type dialog box then click **OK**.
3. Enter a unique name into the Program Name on the Terms tab.
4. Enter the following program terms:
 - Occ Limit: \$300 million
 - Reinstatement: 0

Program: Franchise Deductible

Program Details

Program Type: Catastrophe Excess of Loss
 Inception Date: 01/01/2019
 Expiration Date: 12/31/2019
 Modified: 5/6/2019 by air-worldwideU52107

Currency Rate Table: Save Terms in: Display Terms in: Units:
 AIR Default: US Dollar US Dollar Ones

Layers Program Options Relationships EP Adjustments

Terms

Layer	Treaty ID *	Occurrence Limit	Match Loss	Occurrence Retention	Applies to Areas	Applies to Events	Reinstatements	Status
1	ID94840659	300,000,000	No		United States	All	0	submitted

Layer 1

Perils

☒ Earthquake
☒ Tropical Cyclone
☒ Severe Storm
☐ Other Perils:
☒ Earthquake Shake
☒ Wind
☒ Severe Thunderstorm
☐ Inland Flood
☒ Fire Following
☒ Winter Storm
☐ Wildfire/Bushfire
☐ Terrorism
☐ Coastal Flood
☐ Multi-peril Crop
☐ Crop Hail
☐ Pandemic

Triggers

Starting Event #: 1
 Industry Loss Minimum:
 Industry Loss Maximum:
 Company Loss Minimum: 100,000,000.00
 Company Loss Maximum:

Figure 34. Franchise Deductible Program Terms

5. Scroll down to the **Triggers** section and enter a Company Loss Minimum of \$100 million.

Figure 35. Franchise Deductible Triggers

6. Click **Save** to store the newly created program.

Contract Loss Examples

Using the program terms described in the previous section, Touchstone Re produces the contract losses for the company ground-up losses displayed in the figure below.

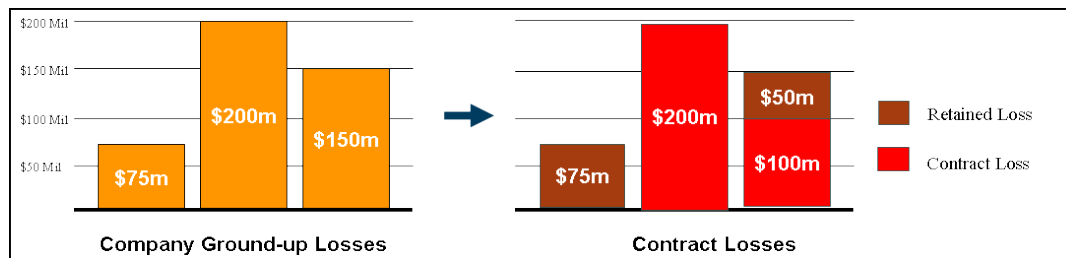


Figure 36. Franchise Deductible Contract Losses

- **Event 1: No contract loss**

This event falls below the trigger criterion (\$100 million); therefore, all loss is retained.

- **Event 2: Contract loss is \$200 M**

This event meets the trigger criterion (\$100 million). Since there is no occurrence retention, the entire \$200 million is eligible contract loss as the occurrence limit is \$300 million. There is no aggregate retention term so the entire loss is paid as a contract loss.

- **Event 3: Contract loss is \$100 M**

The event meets the trigger criterion (\$100 million). The entire \$150 million is eligible as contract loss; however, since there are no reinstatements on the occurrence limit (\$300 million), the contract loss is capped at \$100 million.

11 Reverse Franchise Deductible

A reverse franchise deductible is a per-occurrence loss ceiling that must not be exceeded for a contract loss to be paid. Like a franchise deductible, the ultimate reinsured loss is not reduced by the franchise deductible amount.

Create Per-Event XOL Programs with Reverse Franchise Deductibles

To create a per-event excess of loss (XOL) program with \$300 million occurrence limit and a \$100 million reverse franchise deductible:

Procedure

1. From the all Programs screen, click **New**.
2. Select **Catastrophe Excess of Loss** from the **Add Program** dialog.
3. Enter a unique name into the **Name** field, then click **OK**.
4. Enter the following program terms:
 - **Occurrence Limit:** \$300 million
 - **Reinstatements:** 0

Layer	Treaty ID	Occurrence Limit	Match Loss	Occurrence Retention	Applies to Areas	Applies to Events	Reinstatements	Status	% Placed	Insurer's Concession	Gross Participation	Net Participation	Participation 3	Participation 4	Program Category	Program
1	ID57754280	300,000	No		United States	All	0	Submitted	100%	0%	0%	0%	0%	0%		

Figure 37. Franchise Deductible Program Terms

5. Under the **Triggers** section on the **Layers** tab, enter a **Company Loss Maximum** of \$100 million.

6. Click **Save** to store the newly created program.

Figure 38. Franchise Deductible Triggers

Contract Loss Examples

Using the program terms described in the previous section, Touchstone Re produces the contract losses for the company ground-up losses displayed in the figure.

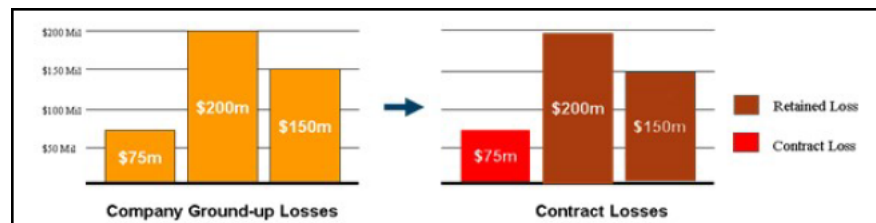


Figure 39. Reverse Franchise Deductible Contract Losses

- **Event 1: Contract loss is \$75M**

This event falls below the trigger criterion (\$100 million). Since there is no occurrence retention and the deductible trigger does not affect reinsured loss, the full \$75m is eligible as contract loss.

- **Event 2: No Contract Loss**

This event exceeds the trigger criterion (\$100 million); therefore, all loss is retained.

- **Event 3: Contract loss is \$100 M**

As with Event 2, this event exceeds the trigger criterion (\$100 million); therefore, all loss is retained.

12 Cascading Structures

Cascading layers are structured in a way that when the limit of a previous layer is exhausted, the next layer drops down in its place.

These structures can be modeled in Touchstone Re using the Inuring functionality that allows losses outside of the layer (below the retention and above the limit) to feed into other programs and layers.

By inuring each layer to another, the losses from the previous layer will be removed before applying the next set of layering terms.

Creating a Cascading Structure

To create a program with the following three layers that cascade:

- \$20 million in excess of \$4 million
- \$50 million in excess of \$4 million
- \$16 million in excess of \$4 million

Procedure

1. Navigate to the **All Programs** tab.
2. Select  to create a new program.
3. Select **Catastrophe Excess** program from the drop-down.
4. Enter the following terms for the first layer:
 - **Occurrence limit:** \$20 Million
 - **Occurrence retention:** \$4 Million

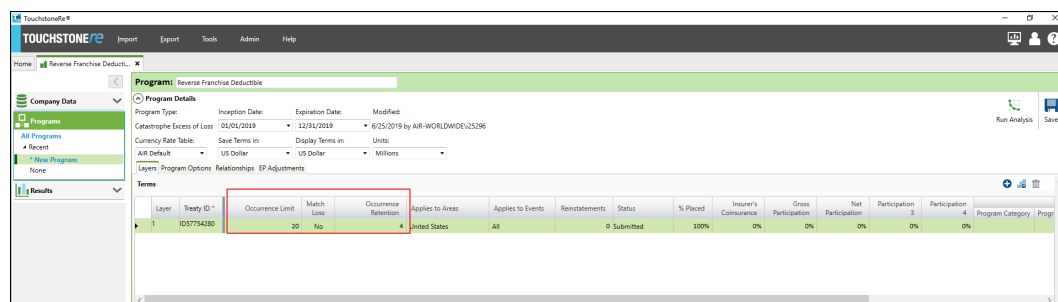


Figure 40. CAT XOL Program for Each Individual Layer

5. Repeat steps 2, 3 and 4 for the other two layers by selecting a **Catastrophe Excess Program** for layer 2 and layer 3 (separately).

6. Click **Save**.

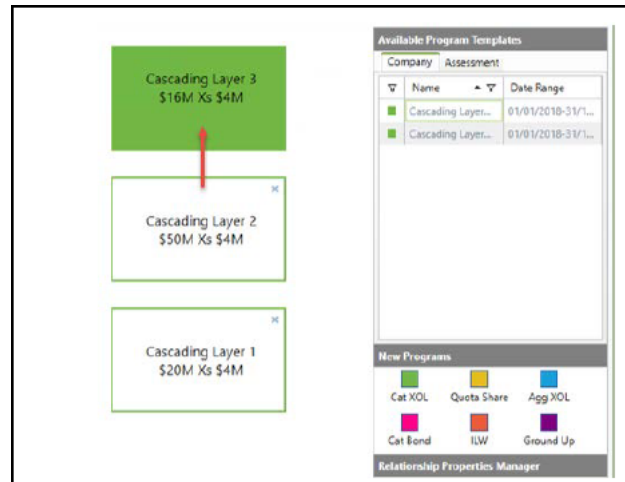


Figure 41. Drag-and-Drop Programs Onto One Another to Define Relationship

7. Within the final layer structure program, select the **Relationships** tab on Layer 3. Within this tab, drag the Layer 2 program on top of the Layer 3.

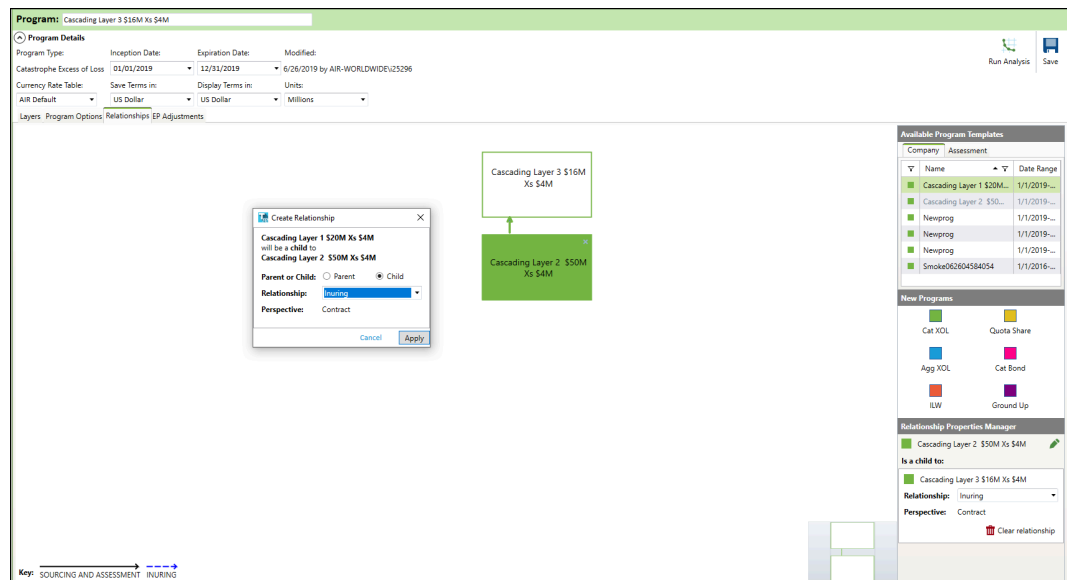


Figure 42. Create Relationship Invoked Once Program Has Been Dropped onto Another

8. Drag layer 1 on top of layer 2 and insure to the benefit of layer 1 by defining its relationship.

9. Select to inure to the benefit of Layer 2 by defining the relationship and perspective.

Note

Each layer has a relationship with another layer, the arrow indicates the flow of loss. The dotted line between layers below indicates losses are inuring.

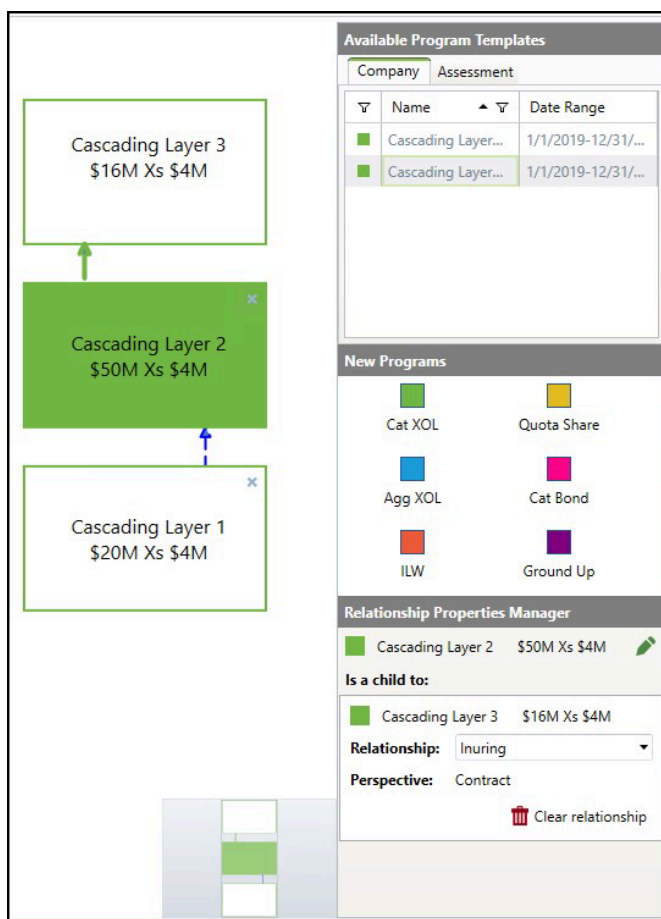


Figure 43. Dotted Line Indicates the Losses Are Inuring

Contract Examples

Using the program terms described in the previous section, Touchstone Re produces the contract losses for the company ground-up losses displayed in the figure below. An Example event loss of \$100 million has been used;

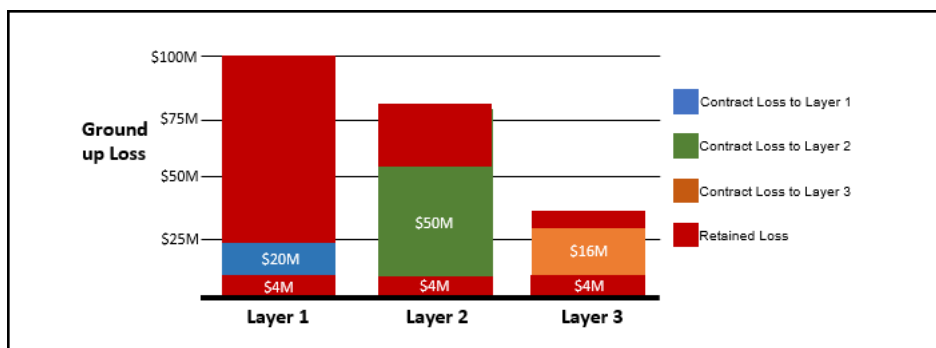


Figure 44. Ground Up and Contract Loss for Each Layer Within Cascading Structure

- **Layer 1: \$20 million contract loss**

The retention of the first layer is applied to the event of \$100 million, reducing this loss to \$96 million. This loss is limited to \$20 million after applying layer 1 occurrence limit, therefore Layer 1 is eligible to \$20 million contract loss.

- **Layer 2: \$50 million contract loss**

As the loss feeds into layer 2, the loss from layer 1 is removed before the terms of layer 2 are applied. The loss feeding into layer 2 is \$100 million - \$20 million which leaves a ground up loss of \$80 million entering layer 2.

After applying the occurrence retention of \$4 million, this reduces the loss to \$76 million. The limit of layer 2 is applied so that \$50 million is eligible for contract loss for layer 2.

- **Layer 3: \$16 million contract loss**

As the loss feeds into layer 3, the loss from layer 2 and 1 is removed before the terms of layer 3 are applied. The loss feeding into layer 3 is \$100 - \$50 - \$20 which leaves a ground up loss of \$30 million entering layer 3.

After applying the occurrence retention of \$4 million, this reduces the loss to \$26 million. This loss is limited to \$16 million after applying the layer 3 occurrence limit. Layer 3 is eligible to \$16 million contract loss.

13 Top-and-Drop

In Touchstone Re a *Top-and-Drop* structure is a style of layering programs rather than an actual program type. It provides the flexibility to reuse the top excess-of-loss layer in the reinsurance tower if the retention of this top layer is not breached by the first loss event. If the layer remains intact it acts as a reinstatement on one or more of the lower layers.

Depending on how the layer operates it can be modeled similarly to the structures described above.

Top-and-Drop Using Step Layers

If the Top-and-Drop layer is to be used as both a top cover and can subsequently reinstate as a lower layer after other layers have been exhausted, it can be modeled as a step layer. This means that the first layer must be exhausted before the layer drops down to provide cover at a lower retention.

See [Section 5](#) which discusses how to model as a drop-down cover.

Top-and-Drop Modeled as a Cascading Structure

The Top-and-Drop can alternatively be modeled in a similar way to the cascading structure described in a chapter later in this document. The lower CAT XOL layers inure to the benefit of the Top-and-Drop layer, meaning that the Top-and-Drop will attach either in the case of a single large event which exhausts the limit of the lower XOL layers, or in a subsequent event if the rest of the program is exhausted.

The possible limitation in modeling the Top-and-Drop layer this way is that if the first event is large enough, the occurrence retention will be applied twice which may not be required. In this case use an additional program to remove the retention, with the net retained company loss then becoming the source for the Cat and Top-and-Drop programs.

Create a Top-and-Drop Structure

To create a program with the following Top-and-Drop structure:

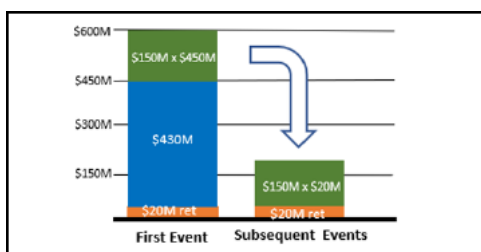


Figure 45. Program Structure Showing Top-and-Drop Layer Structure

- Layer 1: 80M Xs 20M
- Layer 2: 100M Xs 100M

- Layer 3: 250M Xs 200M
- Top-and-Drop Layer: 150M Xs 450M or 150M Xs 20M (subsequent event)
- No reinstatements on any layers

Procedure

1. First create a Catastrophe XOL layer for the cedant retained loss.
2. Navigate to **All Programs** tab.
3. From the **All Programs** tab, select **New**.

The Add Program dialog opens.

4. Select **Catastrophe Excess of Loss**, name the program and then click **OK**.
5. Enter the following program terms on the **Layers** tab:
 - **Occurrence Limit:** \$20M
 - **Reinstatements:** 99 (unlimited)
 - **Occurrence Retention:** \$0M

Figure 46. Top-and-Drop Program

6. Follow the same steps to create a new program for the **Catastrophe XOL** layers with occurrence limits and retentions as follows:
- **Layer 1:** \$80M Xs \$0M
 - **Layer 2:** \$100M Xs 80M
 - **Layer 3:** 250M Xs \$180M

The screenshot shows the Touchstone software interface for configuring a CAT Layers Program. The main area displays the 'Program Details' for 'CAT Layers'. The 'Terms' table lists three layers with their respective occurrence limits and retentions. The bottom section shows the 'Layer 1' details, including a list of perils and triggers.

Layer	Steady ID	Occurrence Limit	Match Loss	Occurrence Retention	Applies to Areas	Applies to Events	Reinstatements	Status	% Placed	Insurer's Consequence	Gross Participation	Net Participation	Participation 3	Participation 4	Program Category
1	1096753036	80	No	0	United States	All	0	Submitted	100%	0%	0%	0%	0%	0%	
2	1096753037	100	No	80	United States	All	0	Submitted	100%	0%	0%	0%	0%	0%	
3	1074432621	250	No	180	United States	All	0	Submitted	100%	0%	0%	0%	0%	0%	

Figure 47. CAT Layers Program

7. Follow the steps above to create a new program for the Top-and-Drop Cat Layer with:
- **Layer 1:** \$150M xs \$0
8. Click **Save**.
9. Select the **Relationships** tab to create the appropriate **Inuring** relationships.

10. Open the Top-and-Drop program and navigate to the **Relationships** tab.

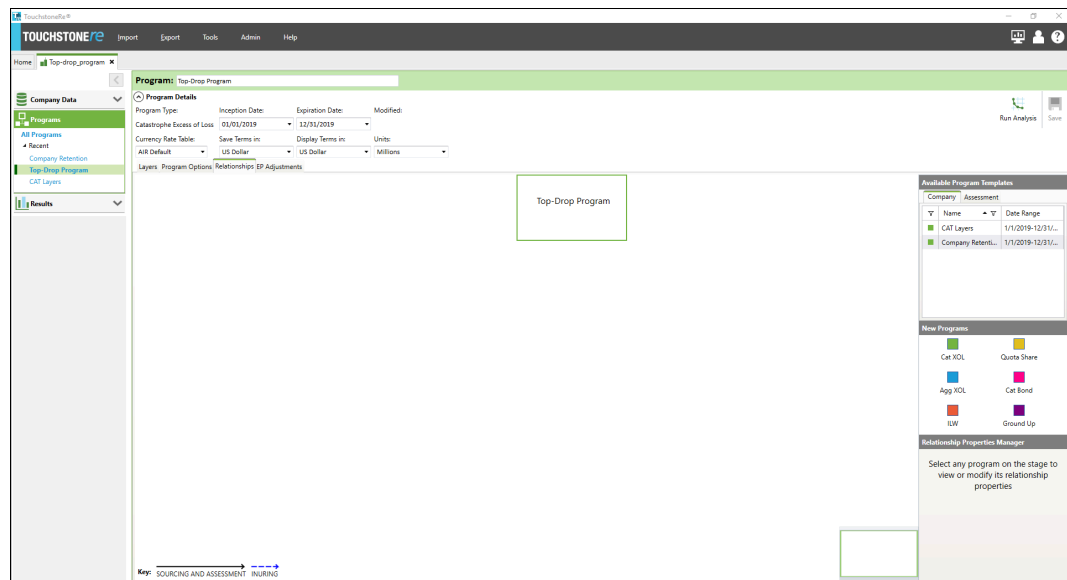


Figure 48. Parent Program (Top or Drop Layer) and Layers to be Sourced and/or Inured

The Top-and-Drop layer is seen in the main window and to the right are the layers which can be related to this.

11. Drag the **Cat Layers** program from the right-hand window on top of the **Top-and-Drop**.

The **Create Relationship** dialog box opens.

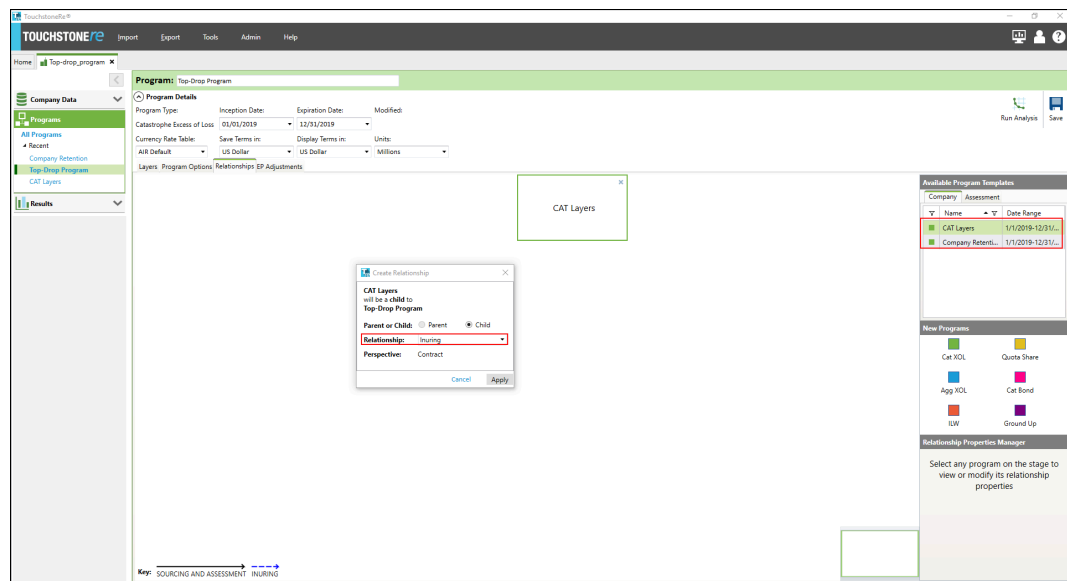


Figure 49. Drag-and-Drop Program on top of another in the Relationships Pane to Invoke Create Relationship

12. Select the relationship and click **Apply**.

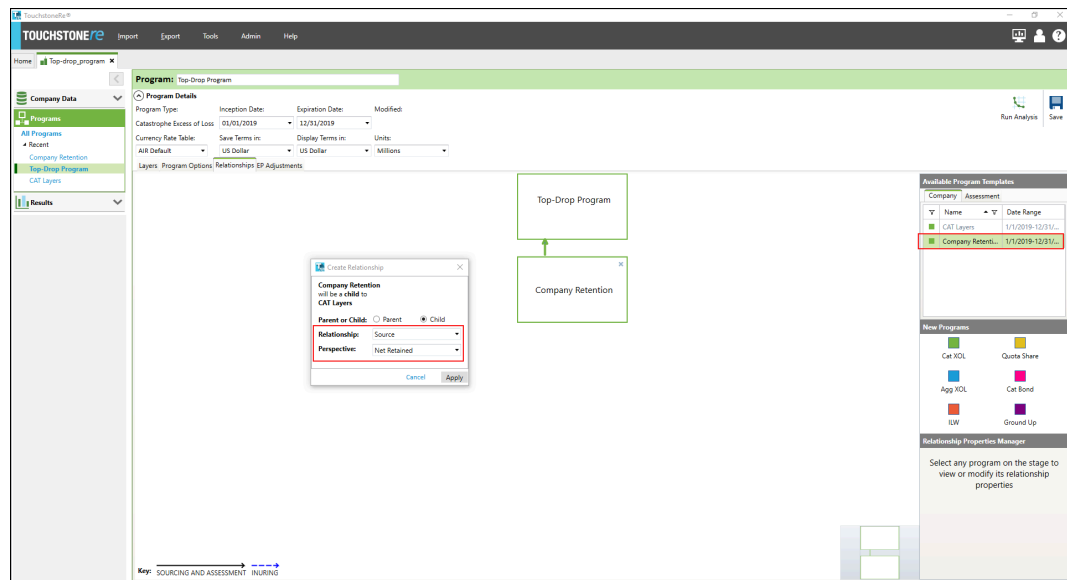


Figure 50. Source Company Retention Layer: Drag-and-Drop Over the Catastrophe Layers

13. Drag the **Company Retention** program from the right-hand window on top of the **Cat Layers** program and select the relationship.

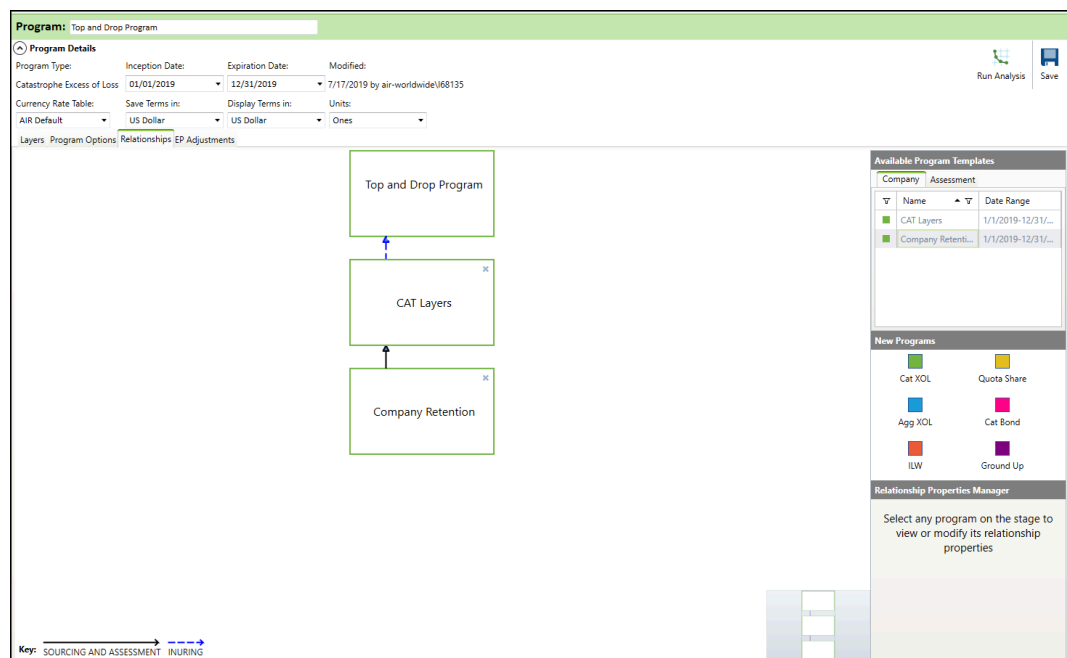


Figure 51. Three-Tier Structure with Company Retention
Sourcing and Cat Layers Inuring to Top-and-Drop Program

14. Click **Save**.

After setting up relationships, navigate to **Program Options** for the Company Retention program and then associate an Exposure Set.

Contract Loss Examples

Using the program terms described in the previous section, Touchstone Re produces the contract losses for the company ground-up losses displayed in the figures below.

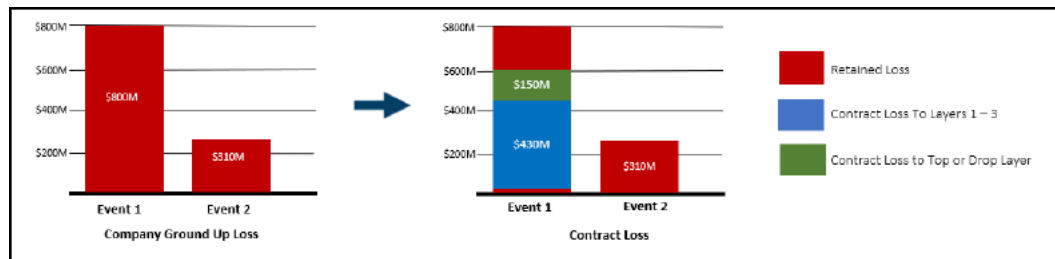


Figure 52. Event Losses Flow Through Program Where First Event is Large Enough to Reach Limit of Top Layer

- **Event 1:** Contract Loss of \$150m to Top-and-Drop layer

The retention program removes \$20m of the \$800M event loss with the remaining event loss of \$780M sourcing into the cat program. The first \$430m of loss is absorbed by the CAT XOL layers. This is then also removed from the total loss that feeds into the top layer. The loss feeding into the Top and Drop is \$350m which is capped at the layer limit of \$150M.

- **Event 2:** No contract loss

The full CAT XOL program, including the Top-and-Drop has been exhausted.

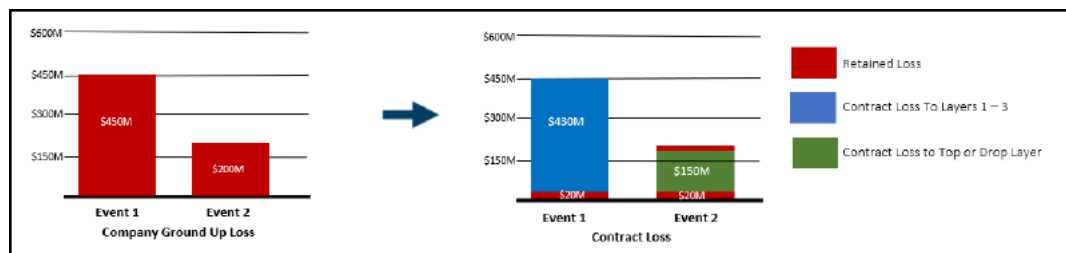


Figure 53. Event Losses Flow Through Program Where First Event Does Not Reach Program Limits So Top Layer Drops Down for Second Event

- **Event 1:** No Contract Loss to Top-and-Drop layer

\$20m of loss is removed by the retention program, with the remaining \$430 sourcing into the CAT XOL program which provides cover up to the limit. Inuring this to the benefit of the Top-and-Drop layer removes all remaining loss with nothing feeding into the top layer.

- **Event 2:** Contract Loss of \$200m to Top-and-Drop layer

The retention program removes \$20m of loss and the remaining event loss sources into the Catastrophe program. The lower CAT XOL layers have been exhausted but the top layer drops down.

The Top-and-Drop program takes loss up to the limit of \$150m. There is an additional \$30m of loss retained by the cedant.

14 Nth-Event Covers

“Nth”-event cover contracts (where “N” stands for any number greater than 1) provide coverage above a per occurrence retention when a company is willing to absorb the loss above the retention up to some occurrence limit for the first N number of occurrence limits. Second event covers, for example, would not pay for the first event producing loss that exceeds the occurrence retention. The Nth event may be defined as an event that exceeds the occurrence retention or as the event occurring when the full occurrence limit has been exhausted (similar to a contract with an amount otherwise recoverable where the amount is the occurrence limit multiplied by N).

Creating Nth Event Covers

There are two different methods for creating Nth event covers in Touchstone Re: using an Nth occurrence limit (otherwise recoverable) and using an Nth event exceeding occurrence retention. This section describes both methods.

Second Occurrence Limit (Otherwise Recoverable)

To create a program with a \$250 million occurrence limit in excess of \$150 million as a second event cover:

Procedure

1. From the Company Data pane, select the **Programs** tab and then click [?](#).
2. Select **Aggregate Excess of Loss** from the **Add Program Type** dialog box and then click **OK**.
3. Enter a unique name into the **Program Name** on the **Terms** tab.

4. Enter the following occurrence and aggregate terms:

- **Occ Limit:** \$250 million
- **Occ Ret:** \$150 million
- **Agg Limit:** \$250 million
- **Agg Ret:** \$250 million

Program: 2nd Event Cover (Occ Limit)

Program Details

Program Type: Inception Date: 01/01/2019 Expiration Date: 12/31/2019 Modified: 5/6/2019 by air-worldwide\j52107

Aggregate Excess of Loss: Currency Rate Table: AIR Default Save Terms in: US Dollar Display Terms in: US Dollar Units: Ones

Layers Program Options Relationships EP Adjustments

Terms

Layer	Treaty ID *	Occurrence Limit	Match Loss	Occurrence Retention	Aggregate Limit	Aggregate Retention	Multi-Year Reset Date
1	ID26393867	250,000,000	No	150,000,000	250,000,000	250,000,000	

Figure 54. Nth Occurrence Limit (Otherwise Recoverable) Program Terms

Note



By setting the Agg Ret equal to the Occ Limit, you ensure the contract will never pay out on the first event because the Aggregate Retention will always be greater than or equal to the loss eligible to be paid out for the event. In cases where the Aggregate Retention is greater than the eligible occurrence pay out (i.e., if the event caused losses smaller than the occurrence limit), only a portion of the Aggregate Retention will be depleted. The remainder will carry over to the next event, causing some or all of the second event losses to be once again retained. The Aggregate Retention continues to carry over event-to-event until fully depleted

5. Click **Save** to store the newly created program.

Second Event Exceeding Occurrence Retention

To create a program with a \$250 million occurrence limit in excess of \$150 million with a second event cover:

Procedure

1. From the Company Data pane, select the **Programs** tab and then click [New](#).
2. Select **Catastrophe Excess of Loss** from the **Add Program Type** dialog box and then click **OK**.
3. Enter a unique name into the **Program Name** on the **Terms** tab.

4. Enter the following occurrence terms:

- Occ Limit: \$250 million
- Occ Ret: \$150 million

Program: 2nd Event Cover (Occ Ret)

Program Details

Program Type: Inception Date: 01/01/2019 Expiration Date: 12/31/2019 Modified: 5/6/2019 by air-worldwide\52107

Catastrophe Excess of Loss: 01/01/2019 12/31/2019 5/6/2019 by air-worldwide\52107

Currency Rate Table: Save Terms in: US Dollar Display Terms in: US Dollar Units: Ones

AIR Default: US Dollar

Layers Program Options Relationships EP Adjustments

Terms

Layer	Treaty ID *	Occurrence Limit	Match Loss	Occurrence Retention	Applies to Areas	Applies to Events	Reinstatements	Status	% Pl
1	ID43919512	250,000,000	No	150,000,000	United States	All		1 Submitted	

Layer 1

Perils

☒ Earthquake ☒ Tropical Cyclone ☒ Severe Storm ☐ Other Perils:
☒ Earthquake Shake ☒ Wind ☒ Severe Thunderstorm ☐ Inland Flood
☒ Fire Following ☒ Winter Storm ☐ Wildfire/Bushfire
☐ Terrorism
☐ Coastal Flood
☐ Multi-peril Crop
☐ Crop Hail
☐ Pandemic

Triggers

Starting Event #: 2 Industry Loss Minimum: Company Loss Minimum:

Industry Loss Maximum: Company Loss Maximum:

Figure 55. Nth Event Exceeding Occurrence Retention Terms

5. Scroll down to the **Triggers** section and set the Starting Event # to 2.

TOUCHSTONE

Program: New Program

Company Data

Program Details

Program Type: Inception Date: 01/01/2019 Expiration Date: 12/31/2019 Modified: 5/13/2019 by air-worldwide\52396

Catastrophe Excess of Loss: 01/01/2019 12/31/2019 5/13/2019 by air-worldwide\52396

Currency Rate Table: Save Terms in: US Dollar Display Terms in: US Dollar Units: Ones

AIR Default: US Dollar

Layers Program Options Relationships EP Adjustments

Terms

Layer	Treaty ID *	Occurrence Limit	Match Loss	Occurrence Retention	Applies to Areas	Applies to Events	Reinstatements	Status	% Placed	Insurer's Commence	Gross Participation	Net Participation	Participation 3	Participation 4	Program Category 1	Prog
1	ID1543075	250,000,000	No	150,000,000	United States	Same		1 Submitted	100%	0%	0%	0%	0%	0%		

Layer 1

Perils

☒ Earthquake ☒ Tropical Cyclone ☒ Severe Storm ☐ Other Perils:
☒ Earthquake Shake ☒ Wind ☒ Severe Thunderstorm ☐ Inland Flood
☒ Fire Following ☒ Winter Storm ☐ Wildfire/Bushfire
☐ Terrorism
☐ Coastal Flood
☐ Multi-peril Crop
☐ Crop Hail
☐ Pandemic

Triggers

Starting Event #: 2 Industry Loss Minimum: Company Loss Minimum:

Industry Loss Maximum: Company Loss Maximum:

Figure 56. Starting Event Number

6. Click **Save** to store the newly created program.

Contract Loss Examples

Using the program terms described in the Second Occurrence Limit (otherwise recoverable) method in the previous section, Touchstone Re produces the contract losses for the company ground-up losses displayed below.

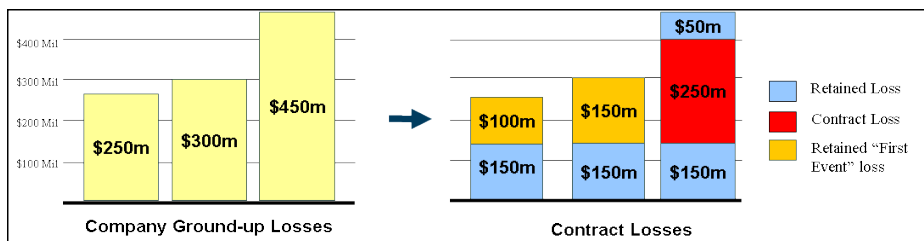


Figure 57. Second Event Cover (Occurrence Limit) Contract Losses

- **Event 1:** No contract loss

The first \$150 million is retained via the occurrence retention. The remaining \$100 million is eligible contract loss; however, it is absorbed in its entirety by the aggregate retention term (\$250 million), depleting the aggregate retention to \$150 million.

- **Event 2:** No contract loss

The first \$150 million is retained via the occurrence retention. The remaining \$150 million is eligible contract loss; however, it is absorbed in its entirety by the aggregate retention term (\$250 million), completely depleting the aggregate retention

- **Event 3:** Contract loss of \$250 M

The first \$150 million is retained via the occurrence retention. Of the remaining \$300 million in loss, only the first \$250 million is eligible contract loss. As the aggregate retention was exhausted on Event 2, the entire \$250 million is paid as contract loss, capping the loss at \$250 million for the year.

Using the program terms described in the Exceeding Occurrence Retention method in the previous section, Touchstone Re produces the contract losses for the company ground-up losses displayed below

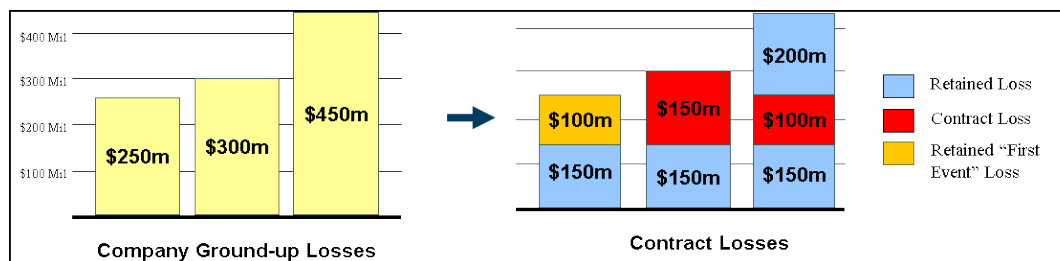


Figure 58. Second Event Cover (Exceeding Occurrence Retention) Contract Losses

- **Event 1:** No contract loss

The first event is excluded from consideration regardless of its size due to the starting event trigger.

- **Event 2:** Contract loss of \$150 M

The first \$150 million is retained via the occurrence retention. The remaining \$150 million is eligible contract loss. Since there are no aggregate terms, the entire \$150 million is paid as contract loss.

- **Event 3:** Contract loss of \$100 M

The first \$150 million is retained via the occurrence retention. Of the remaining \$300 million, \$250 million is eligible as contract loss, however, only the first \$100 million is paid because there is no reinstatement on the occurrence limit for that event and losses are capped at \$250 million for the year.

About AIR Worldwide

AIR Worldwide (AIR) provides risk modeling solutions that make individuals, businesses, and society more resilient to extreme events. In 1987, AIR Worldwide founded the catastrophe modeling industry and today models the risk from natural catastrophes, terrorism, pandemics, casualty catastrophes, and cyber incidents. Insurance, reinsurance, financial, corporate, and government clients rely on AIR's advanced science, software, and consulting services for catastrophe risk management, insurance-linked securities, longevity modeling, site-specific engineering analyses, and agricultural risk management. AIR Worldwide, a Verisk (Nasdaq:VRSK) business, is headquartered in Boston with additional offices in North America, Europe, and Asia. For more information, please visit www.air-worldwide.com.

